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Review Article

Effectiveness of heat-reflective asphalt pavements in mitigating urban heat islands: A systematic literature review

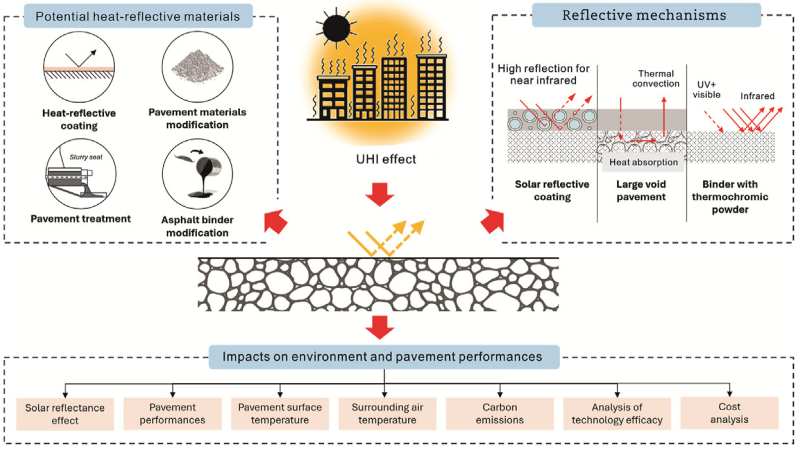
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HIGHLIGHTS

- Cool pavement technologies for mitigation of urban heat island (UHI) are reviewed.
- Pavement materials designed with high solar reflectivity are significant cooling strategies.
- High emissivity and albedo values are preferred for reducing the UHI effect.
- Heat-reflective thermal insulation coatings demonstrated superior cooling performance.
- Heat-reflective pavement is one of the best methods for pavement cooling based on its efficiency and level of readiness.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Heat-reflective pavement
Urban heat island
Cool pavement
Heat-reflective coating
Albedo

ABSTRACT

Urban heat island (UHI) effect is a growing concern in numerous cities worldwide, which increases urban temperatures. Conventional asphalt pavements are a major contributor to the issue, causing environmental and health concerns. To tackle this issue, the implementation of cool pavements such as heat-reflective asphalt pavements has been introduced. This systematic literature review (SLR) thoroughly examines prior research to assess the effectiveness of heat-reflective asphalt pavements in reducing UHI effects. This SLR was conducted in accordance with the preferred reporting items for systematic review and meta-analysis (PRISMA) guidelines to enhance data reliability and minimize bias. This review process involved establishing review protocol, formulating review questions, systematically selecting articles through identification, screening, eligibility, quality appraisal, and data abstraction and analysis from various databases such as Scopus, Web of Science (WoS), Wiley, Taylor Francis, and ScienceDirect. Three primary themes and nine sub-themes were derived from the three review questions. From

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the results, heat-reflective asphalt pavements effectively minimized the UHI effect. However, their efficiency varies depending on factors such as pavement types, paving location, and use of cool materials. A comprehensive analysis examined heat-reflective pavement's mechanisms, benefits, and drawbacks. This investigation aimed to enhance comprehension and establish a robust basis for future studies in this field.

1. Introduction

The urban heat island (UHI) phenomenon, recognized for elevating temperatures and intensifying heatwaves in urban areas, stems from factors like radiative entrapment, heightened heat absorption, inhibited air circulation, reduced greenery, and heightened human activity associated with industrial urbanization (Maria et al., 2013; Synnefa et al., 2011). Solar radiation contributes to elevated ambient temperatures, affecting over 400 major cities globally and raising concerns in the scientific community (Santamouris, 2015). Akbari and Kolokotsa (2016) highlighted the crucial role of pavements in urban thermal equilibrium, influencing the UHI effect through extensive solar absorption. Due to urbanisation and population density, approximately ranging from 50% to 60% of the metropolitan areas consist of roofs and pavement, as reported by Li et al. (2020). According to Takebayashi and Moriyama (2012), the mean temperature of asphalt pavement surfaces was observed 20 °C higher than grass, thereby indicating a detrimental impact of the UHI effect (Zhu et al., 2021). High temperatures in urban areas, a consequence of the UHI effect, pose significant threats to the environment and human well-being. Research indicates that UHI contributes to increased urban pollutants, negatively impacting air quality, and posing health risks (Stathopoulou and Cartalis, 2007; Taha, 2017; Xu et al., 2021). Elevated temperatures also affect thermal comfort and the economy, leading to heightened electricity demand for cooling and reduced demand for heating (EPA, 2021). The heat transfer processes in pavements are illustrated in Fig. 1, including conduction, convection, reflection, and radiation emissions. Solar radiation entering the atmosphere strikes pavements, being reflected, absorbed, and transported through the surface into the ground (Kappou et al., 2022). Heat transmission and pavement temperatures are also influenced by evaporation from rain and surface water, contributing to the complex dynamics of the UHI phenomenon.

Asphalt pavement temperatures, influenced by material thermal characteristics, play a crucial role in urban heat dynamics. Energy transfer from hot pavements to the surrounding air is governed by temperature gradients and convection. The application of cool paving materials has been explored to counteract traditional asphalt's thermal impact, potentially reducing the UHI effect and energy consumption (Croce et al., 2021). Cool pavements, investigated since the late 20th century, aim to lower surface temperatures, and enhance outdoor comfort (Levine, 2011; Wong et al., 2022). Santamouris (2013) suggested that implementing pavement cooling can improve longevity, reduce waste from maintenance, and address issues like rutting and thermal cracking.

Various cooling techniques, including white topping, light aggregates, concrete additives, and reflective synthetic binders, have been traditionally employed (EPA, 2021). Emerging technologies like “cool pavements” are categorised into three types: reflective, evaporative, and heat storage modification (Anupam et al., 2021) as illustrated in Fig. 2. Reflective pavements, characterized by higher solar reflectance and lighter coloration, mitigate the urban heat island (UHI) effect by reflecting incoming solar radiation, reducing energy absorption, and sensible heat emission (Anupam et al., 2022; Rossi et al., 2016). Besides, materials with dark, rough surfaces tend to absorb more solar radiation than cool materials with smooth, light-colored surfaces (Santamouris and Kolokotsa, 2016). Evaporative pavements utilize water for cooling

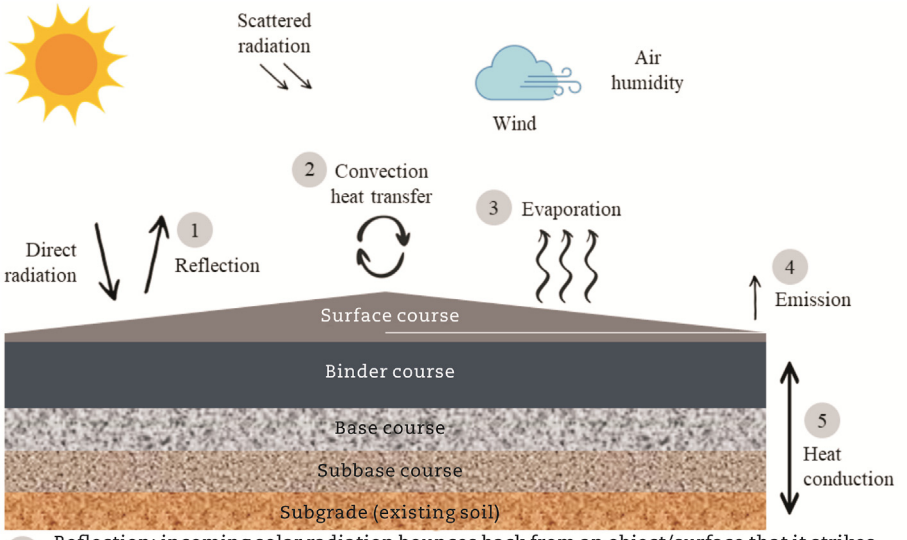


Fig. 1. Heat transfer in pavements (Kappou et al., 2022).

through evaporation, with the cooling effect attributed to the absorption of latent heat during the phase transition of water from liquid-to-vapor, assuming there is a continuous supply of water.

Although numerous studies address types and technologies used for mitigating the UHI effect, there exists a significant gap in comprehending the resilience and effectiveness of a specific category of cool pavement-heat-reflective pavements. This gap pertains specifically to understanding how these pavements respond to weather-induced stresses and how their thermal behaviour influences the overall performance of asphalt pavements. The need for a more in-depth exploration of the impact of environmental conditions on the durability and thermal characteristics of heat-reflective pavements is evident, addressing a crucial aspect in the broader context of UHI mitigation strategies. This review will concentrate primarily on the employed materials and methods, the mechanisms, benefits, and drawbacks, and the cooling efficacy and cost efficiency of heat-reflective asphalt pavement. The present systematic literature review (SLR) is guided by a set of review questions, including but not limited to: (a) what are the materials and techniques that have the potential to be applied in heat-reflective asphalt pavements? (b) what are the mechanisms, benefits, and drawbacks of utilizing heat-reflective asphalt pavements to mitigate UHI effects, and (c) how do the cooling performance and cost-effectiveness of the heat-reflective asphalt pavements influence the reduction of UHI.

Overall, this study aims to evaluate the effectiveness of heat-reflective pavement as a technological approach to alleviate UHI effects based on existing literature review. This study offers significant contributions to researchers and specialists working on pavements and UHIs. The development of cool pavement technologies is instrumental in attaining the targets outlined in the sustainable development goals, including SDG 9 (industry, innovation, and infrastructure), SDG 11 (sustainable cities and communities), and SDG 13 (climate action). The UHI effect will eventually lead to climate change, posing significant challenges, impacting global necessities such as fresh water, food security, and energy for a significant portion of the global population. The least developed and underdeveloped nations are expected to experience the most severe impacts and possess the least resources to cope with the projected disruptions to their social, economic, and environmental surroundings. Thus, this review serves as a basis for developing impactful public policies to address climate change and UHIs.

2. Methodology

2.1. Review protocol–PRISMA 2020

In ensuring an unbiased approach to data selection and analysis, a systematic literature review requires the establishment of a review protocol (Kitchenham and Brereton, 2013). This study is scoping based on the latest Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA Statement) published in 2021 (PRISMA, 2021). PRISMA, dividing into 7 sections, consists of 27 items that cover all aspects of the review, including abstract, objectives, research questions, inclusion criteria, search strategies, quality assessment criteria and screening procedures, data extraction and synthesis, as well as reporting strategies (Gomersall et al., 2015; Page et al., 2021). The present protocol begins by outlining the review questions, followed by an explanation of the search strategies, encompassing three subprocesses: identification, screening, eligibility, and inclusion. The establishment of inclusion and exclusion criteria enables a systematic screening process of primary studies identified through the employed searching strategy (Templier and Paré, 2015). Then, the authors assessed the selected papers by scrutinizing their complete contents during the quality appraisal process. Subsequently, the data pieces obtained from the primary studies are identified to address the review questions. After data extraction, a thorough analysis and synthesis are conducted to draw conclusions based on review questions.

2.2. Formulation of research questions (RQs)

The process of conducting literature reviews involve research inquiries, which necessitate the guidance of research questions throughout the various stages of inquiry (selected studies, data extraction, synthesis methodology, and report method). Therefore, the PRISMA checklist has been modified to develop the review questions. The present SLR formulates the research questions based on identified knowledge gaps. Cronin et al. (2008) emphasized that the importance of formulating precise research questions to limit the scope of the review and facilitate the identification of a feasible amount of data. Extensive research has been conducted on the various technologies used on the pavement to mitigate the effect of UHI. The efficacy of these developed technologies in reducing heat and their cost-effectiveness has also been thoroughly examined. However, the previous literature on the subject matter is restricted in its scope, as it fails to comprehensively investigate the durability and performance of heat-reflective pavement in mitigating the deleterious effects of environmental pollution. Therefore, this study will focus on the knowledge gap to formulate the three review questions.

2.3. Systematic review process in article selection

The SLR employs the PRISMA flow diagram as a search framework, comprising four steps: identification, screening, eligibility, and inclusion (PRISMA, 2021). The initial search, as indicated by the PRISMA flow diagram, entails the examination of several databases. However, for this SLR, the final articles included are solely derived from two main databases, Scopus and Web of Science (WoS), along with other databases,

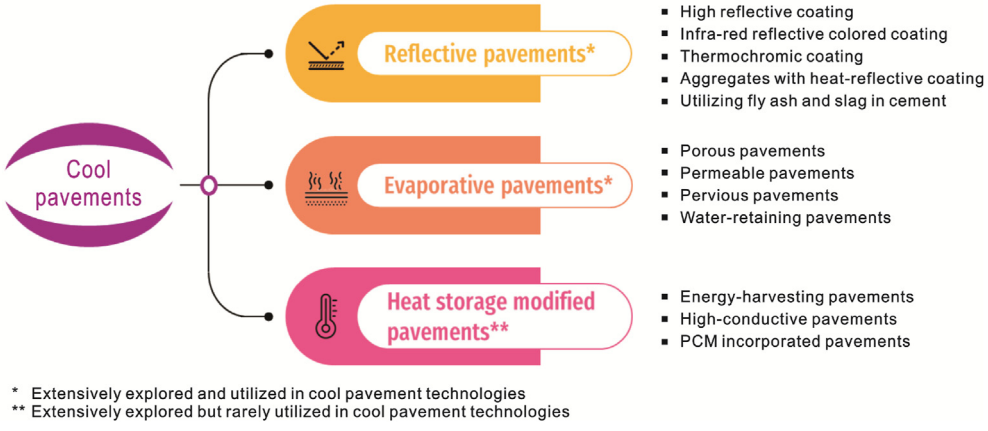


Fig. 2. Cool pavement technologies (Anupam et al., 2021).

namely Wiley, Taylor Francis and ScienceDirect. This SLR consists of four main steps in the systematic searching techniques, including identification, screening, eligibility, and inclusion as referred to Fig. 3.

2.3.1. Identification

Identification is the primary step in the PRISMA protocol, which is a process to search for any synonym, related terms, and variation for the selected keywords. The keywords are developed based on the three review questions of this SLR. The keyword identification is facilitated by utilizing the concept domain and other significant elements (Kitchenham and Brereton, 2013). Rowley and Slack (2004) suggested that incorporating synonyms, abbreviations, alternative spellings, and other related terms can broaden the scope of the search statement. Table 1 shows the main keywords and search strings derived from the three review questions of this SLR. The proposed terms were enriched by the online power thesaurus, keywords in previous research, and a variation of keywords suggested by Scopus. The authors developed a search string using the terms through phrase searching, field codes, Boolean operators, truncation, and wild card in both main journal databases, which are Scopus and WoS, and other databases such as Wiley, Taylor Francis, and ScienceDirect (Brereton et al., 2007; Flemming, 2010). Scopus is one of the biggest abstracts and citation databases of peer-reviewed literature which covers 82.4 million entries and 36,377 titles from 11,768 publishers worldwide. WoS is a comprehensive research database and citation index that consisting of more than 33,000 journals across 256 disciplines that includes a broad scope covering natural sciences, social issues, and development and planning.

According to Kitchenham and Brereton (2013), Boolean operators (such as AND, OR, or NOT) can effectively limit the scope of a subject by merging the key concepts and their corresponding synonyms. The search strings were developed based on each review question and applied in the databases to search solely within the articles' title, abstract, and keyword. The searching process in the databases resulted in a total of 571 articles for the three review questions as demonstrated in Fig. 3.

2.3.2. Screening

During the screening stage, the duplicates as well as the inclusion and exclusion criteria filter out irrelevant articles from the 571 articles reviewed in the identification stage. First, the duplicates are eliminated by checking the repeated article title after the identification stage and the remaining 270 articles will be filter based on the inclusion criteria. The screening stage is restricted to articles published within the past decade

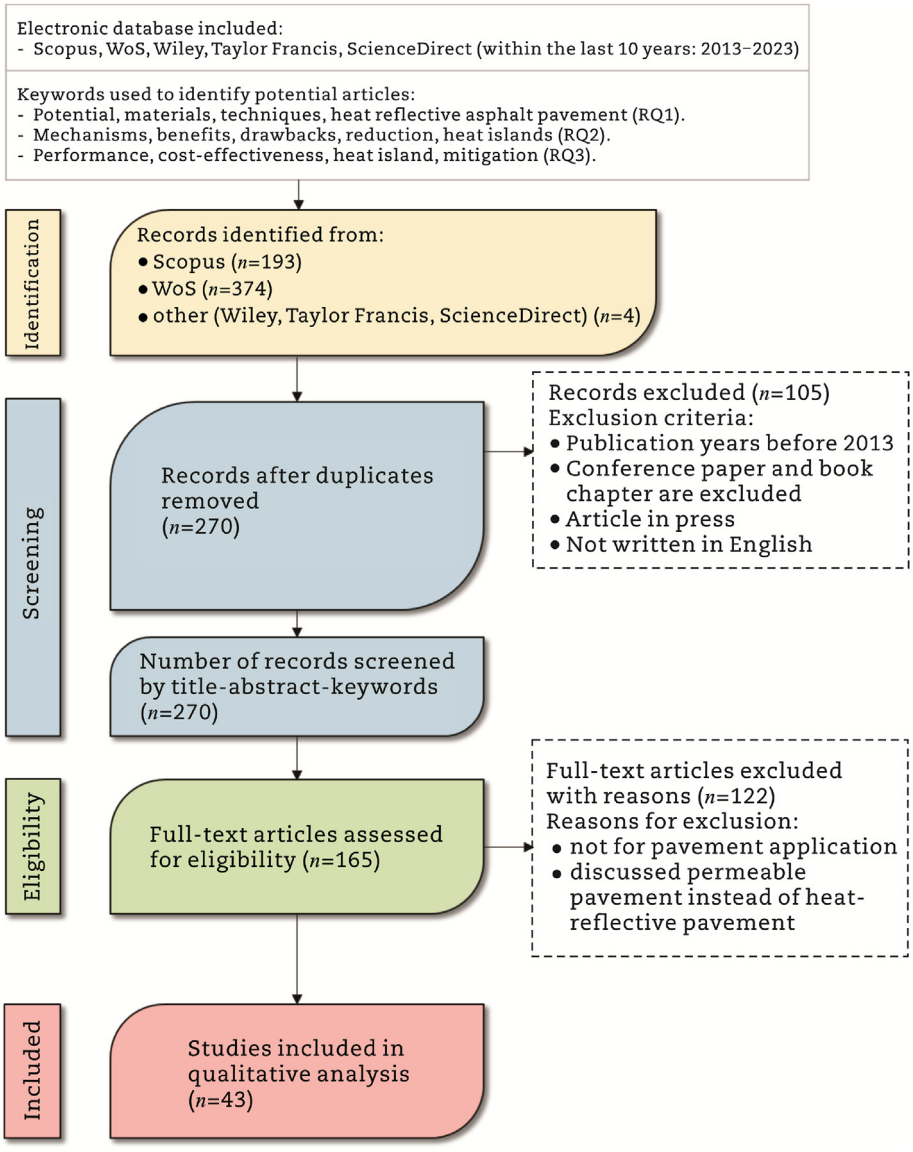


Fig. 3. Flow diagram of the retrieved articles in systematic searching strategies.

Table 1
Applied keywords and search strings derived from the research questions.

RQ	Keyword	Search string
RQ1	1. Potential 2. Materials 3. Techniques 4. Heat-reflective asphalt pavements	(potential* OR possibl* OR prospect* OR probable OR developing) AND (material* OR matter* OR substanc* OR stuff* OR medium*) AND (technique* OR method* OR approach* OR procedure* OR process* OR system* OR way* OR manner* OR mode* OR style* OR strateg* OR practic*) AND (“heat reflective pavement*" OR "solar reflective pavement*" OR "cool pavement*")
RQ2	1. Mechanisms 2. Benefits 3. Drawbacks 4. Heat-reflective asphalt pavements 5. Reduction 6. Heat islands	(mechanism* OR procedur* OR process* OR system* OR operation* OR method* OR technique*) AND (“heat reflective pavement*" OR "solar reflective pavement*" OR "cool pavement*") AND (reduction* OR depletion* OR mitigation* OR limit* OR "ease* OR alleviation* OR lowering”) AND (“heat island*" OR "urban heat island* effect*" OR "urban heat island*" OR "heat island* effect*") (benefit* OR good OR satisfaction* OR advantag* OR contribution* OR merit* OR strength* OR usefulness OR improvement* OR value* OR "positive outcom*") AND (“heat reflective pavement*" OR "solar reflective pavement*" OR "cool pavement*") AND (reduction* OR depletion* OR mitigation* OR limit* OR "ease* OR alleviation* OR lowering”) AND (“heat island*" OR "urban heat island* effect*" OR "urban heat island*" OR "heat island* effect*") (drawback* OR disadvantag* OR shortcoming* OR limitation* OR issue* OR hindranc* OR defect* OR downside* OR deficienc* OR obstacl* OR imperfection* OR constraint*) AND (“heat reflective pavement*" OR "solar reflective pavement*" OR "cool pavement*") AND (reduction* OR depletion* OR mitigation* OR limit* OR "ease* OR alleviation* OR lowering”) AND (“heat island*" OR "urban heat island* effect*" OR "urban heat island*" OR "heat island* effect*")
RQ3	1. Performance 2. Cost-effectiveness 3. Heat-reflective asphalt pavements 4. Heat island 5. Mitigation	(performanc* OR accomplishment* OR achievement* OR efficienc* OR outcom*) AND (“heat reflective pavement*" OR "solar reflective pavement*" OR "cool asphalt pavement*") (“cost-effectiveness” OR "cost-effective” OR "cost-benefit” OR "cost-efficient” OR "cost-based” OR "cost-cutting” OR "cost-effectively” OR "low-cost” OR "economical” OR "customer-friendly”) AND (“heat reflective pavement*" OR "solar reflective pavement*" OR "cool pavement*")

(2013–2023). The timeline is selected such that sufficient articles are available for a thorough and qualitative analysis. In addition to that, the available document categories are restricted to definitive published articles and review papers from various journals. The content of journal articles is of greater quality than conference papers since they are subject to peer review. The authors only included those written in English to avoid confusion and translation difficulties in this SLR. After applying the aforementioned criteria for string filtering, 105 articles were excluded due to non-compliance with the inclusion requirements. The eligibility process was applied to the remaining 165 articles.

2.3.3. Eligibility

The authors conducted an eligibility assessment process based on their titles and abstracts to ensure that the remaining articles adhered to the pre-established inclusion and exclusion criteria. Irrelevant research will be excluded, and the remaining studies will be evaluated in this study (Xiao and Watson, 2019). According to Moola et al. (2015), two important steps should be taken during the final evaluation process. During the initial stage, a researcher will peruse the article's title and abstract, along with the introduction and conclusion sections, if applicable, to eliminate any potentially misleading articles generated by search engine. Subsequently, a researcher will ascertain whether essential information is necessary to extract data. Any papers that are deemed irrelevant or inaccessible will be excluded from consideration. In total, 122 articles were excluded after the authors reviewed the title and abstract. The eliminated articles did not concentrate on the heat islands effect; instead, they highlighted alternative cooling techniques rather than the heat-reflective asphalt paving techniques. Additionally, the articles centred on the consequences of the heat islands rather than their potential solutions and benefits. There are 43 papers that are yet to undergo the quality appraisal process prior to undergo data extraction and detail analyzation.

2.4. Quality appraisal

It is recommended that researchers acquire complete texts of studies for quality evaluation after screening for inclusion, as suggested by Xiao and Watson (2019) and Carroll et al. (2013). Conducting a quality appraisal is crucial for evaluating the underlying reliability of a study, as it helps to identify and eliminate any potential methodological biases that may be present in the study (Ludvigsen et al., 2016; Petticrew and Roberts, 2008). The eligible articles will undergo a comprehensive examination, involving the data collection methodology, data analysis techniques, results, and conclusions prior to the data extraction and synthesis phase (Flemming, 2010).

A checklist is necessary to be developed to conduct qualitative and quantitative analysis of the selected articles (Neely et al., 2014). In this SLR, the checklist framework developed by Dieckmann et al. (2009) is utilized and modified to be more applicable to the present review. The checklist has 18 items, and each of the 43 papers will be scrutinised to verify that it fits the requirements for each item. The remaining articles are categorized as either low or high quality, with those that satisfy the criterion of item 15 out of 18 (80%) being deemed good quality. In this process, all the papers were considered high quality since they all satisfied the standards by a minimum of 90%. Therefore, every article was eligible for evaluation.

2.5. Data abstraction and analysis

Data abstraction aims to extract the essential information from each article by examining it extensively and identifying the pertinent content so that it can be methodically structured and linkages to the review questions can be created (Arnold and Fletcher, 2012; Harden and Thomas, 2005). Techniques such as mixed-method or qualitative are often used to abstract data by combining charts, tables, and textual descriptions (Kastner et al., 2012; Xiao and Watson, 2019).

This SLR used a segregated design of mixed method review. It entails synthesizing qualitative and quantitative studies independently in accordance with their synthesis traditions and merging the findings textually (Sandelowski and Leeman, 2012). The integration of quantitative analyses was achieved through using meta-analyses. In contrast, the analysis of qualitative data involved the identification of descriptive themes, which were subsequently distilled into analytical themes. The analytical themes offered a framework for synthesizing of the quantitative research outcomes. The abstracted data were collected based on the review questions to ensure the reviewed articles meet the requirements. The authors gathered information from the articles and compiled comparable or related data to form themes and sub-themes. Nine sub-topics emerged from the main themes based on the review questions. The corresponding author and co-authors designated the naming of themes and sub-themes. In developing the theme names, the researcher engaged in discussion and established a consensus to ensure that the names were acceptable and related to the results of the articles.

3. Results and discussions

3.1. Background of the selected articles

There are 43 selected articles included in this systematic literature review. A segregated design of mixed method review, which consists of both qualitative and quantitative studies. The present study has identified and examined three main themes, namely, the identification of prospective materials and methodologies for the development of heat-reflective asphalt pavements, the underlying mechanisms and impacts of heat-reflective asphalt pavements, and efficacy of heat-reflective asphalt pavements in mitigating UHI effects. Further analysis of the main themes has resulted in nine sub-themes. The date range of this SLR is limited to the last ten years (2013–2023) to have the latest and sufficient articles for a thorough and qualitative analysis. This topic only started to get some interest from researchers since 2015, and then it recorded the highest contribution in 2020 due to its effectiveness and superior performance (Fig. 4). However, the attention to the topic has been reduced after 2022. The existing research on the heat-reflective pavement is insufficient and not thoroughly discussed, especially on its durability, serviceability, and mechanical performance on asphalt mixtures. Therefore, it is imperative to conduct a review that provides a comprehensive overview of the technologies, mechanisms, benefits, and drawbacks associated with heat-reflective pavement.

Out of 43 articles surveyed, 30 of them emphasise the application of heat-reflective coating. The second most prevalent approach involves pavement material modification, with 13 articles recorded. Most researchers prefer in-laboratory tests such as prototype installation (Guntor et al., 2014) and simulation in the laboratory by using iodine tungsten lamps for sunlight exposure (Chen et al., 2019; Hu et al., 2021; Sha et al., 2017) whereas some researchers conducted field experiments (Benrazavi et al., 2016; Chen et al., 2022; Kyriakodis and Santamouris, 2018). A combination method was also used by researchers to compare and obtain results with higher accuracy (Li et al., 2020; Zheng et al., 2015). Out of the total of 43 articles, the majority of 27 articles were conducted in Asia (20 articles from China, 4 articles from Malaysia, 1 article from Turkey, Japan, and Saudi Arabia, respectively). In addition, other 9 articles were conducted in America (all from the United States), and 7 articles were conducted in Europe (5 articles from Italy, 2 articles from Greece). Fig. 5 shows the general distribution (number of articles) of the selected articles based on their research years, application methods, experimental setup, and research origins. Table 2 tabulated the summarized main extracted data from the 43 selected articles.

3.2. Developed themes and sub-themes

3.2.1. Potential materials and techniques for heat-reflective asphalt pavements

The recognition of albedo is essential when discussing solar reflectance pavement. The albedo, which refers to the amount of solar radiation reflected by a given surface serves as a fundamental predictor of a material's maximum surface temperature. It is a crucial component in decreasing pavement temperatures in cool pavement technology (Jiang et al., 2020). Therefore, cool pavements constructed with high albedo materials may minimize pavement temperatures, and the heat energy radiated into the atmosphere. However, the non-reflected radiation will be absorbed and permeate into the materials, resulting in asphalt pavements with high temperatures. Other techniques, such as paving materials modification, may be alternatives in altering the pavement temperatures to mitigate the UHI effect. In addition, the surface area and absorption capacity of the materials also affect the radiation absorption rate (Jia et al., 2023).

3.2.1.1. Heat-reflective coating. The common method for producing heat-reflective asphalt pavement involves using heat-reflective coatings. This technique increases the reflectivity of the pavement surface, known as albedo, to achieve desired reflectance property. A heat-reflective coating primarily comprises a resin binder, functional pigments, fillers and additives (Jiang et al., 2020). The resin binder serves as the main constituent material, holding all the ingredients together to form the film, and provides structural integrity to the coating (Cao et al., 2017). This binding action ensures that the heat-reflective coating adheres well to the pavement surface, creating a uniform and durable layer that contributes to its overall effectiveness in reflecting heat and solar radiation.

Epoxy resin is commonly selected by researchers to serve as the solvent-based binder among all matrix materials. It functions as an adhesive by bonding the surfaces of two dissimilar components. Epoxy resins, which form a bond between two surfaces of distinct components, serve as common adhesives due to their flexibility to attach to various materials. Previously (Meng et al., 2023), Yi et al. (2019) incorporated high-viscosity polyurethane-modified epoxy resin with an amine agent added and high-elasticity epoxy resin as the binders. After that, Yi et al. (2021) adopted various formulations of epoxy resins in conjunction with film-forming materials to produce heat-reflective coatings. These formulations included EP (100% epoxy resin), B-A (70% epoxy resin and 30% A-type polyurethane-modified epoxy resin), B-B (70% epoxy resin and 30% B-type polyurethane-modified epoxy resin), and B-C (70% epoxy resin and 30% C-type polyurethane-modified epoxy resin). Jiang et al. (2020) observed that 40% epoxy resin was the optimal incorporation of the curing agent when producing the basic polymer from epoxy resin and polyamide curing agents.

Functional material and filler refer to a powder whose main goal is to increase the heat-reflective coating's reflectivity and cooling performance (Yi et al., 2021). Pigments were sometimes added to the filler to enhance the colour of the coating (Yi et al., 2019). Titanium dioxide (TiO_2) is commonly utilized as the primary functional filler due to its high refractive index, high light scattering capacity, substantial UV shielding, robust stability and oxidation resistance, high reflectance ratio towards near-infrared radiation, and significant shielding of solar thermal radiation (Chen et al., 2022). The excellent cooling effect of TiO_2 particles, ranging in size from 0.2 to 1.0 μm, is attributed to their effective utilization of solar radiation energy concentrated in the visible region (0.40–0.72 μm) and near-infrared light region (0.72–2.50 μm). The distribution of TiO_2 within this range serves to characterize its heat-reflective ability, thus enhancing its cooling properties (Cao et al., 2017). The cooling characteristics also exhibit an 80% increase with the elevation of TiO_2 content from 2% to 14%. Yet, it is exceedingly shiny, suggesting that its usage on road surfaces may compromise driving safety (Li et al., 2020). The glossiness of the titanium-based heat-reflective coatings may be reduced by adding silica, and the reflective filler can be created from a blend of silicon oxide (SiO_2) and titanium dioxide (TiO_2). According to Yi et al. (2021), the optimum ratio for the TiO_2-SiO_2 mixture is 1:1, as presented by the superior cooling performance and

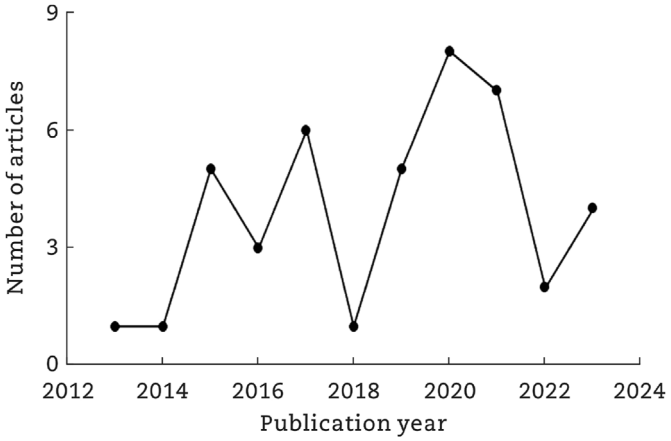


Fig. 4. Number of selected articles with their respective publication years.

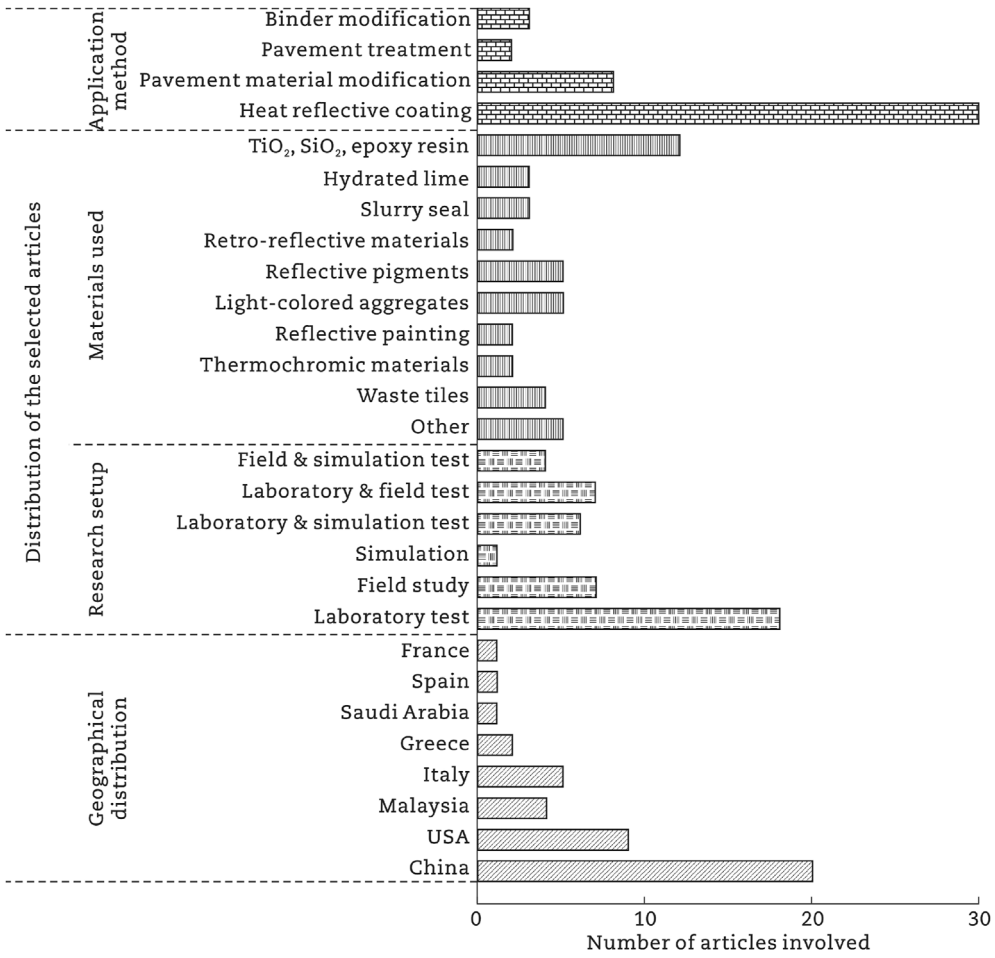


Fig. 5. General distribution of the selected articles.

lower glossiness. Additionally, Zheng et al. (2020) combined floating TiO_2 beads with colour pigments to create their functional material, whereas Chen et al. (2022) used a lower-cost coating with diatomite and hollow glass beads to replace a portion of the expensive TiO_2 to achieve a comparable cooling function. Moreover, the incorporation of low-cost and chemically stable titanium dioxide (TiO_2) with added quantum dots (QDs) results in enhanced quantum yields and increased reflectivity, reaching up to 22%. This modification also imparts fluorescent properties when the inclusion of quantum dots ranges from 5% to 30% (Hu et al., 2023).

The incorporation of additives can enhance both the performance and properties of the heat-reflective coating composition (Li et al., 2016). Jiang et al. (2020) discovered that the additives, which have a considerable effect on the heat-reflective coatings' surface smoothness, durability, storage and construction stability, as well as wetting and dispersion properties, may also act as supplementary film-forming ingredients for the heat-reflective coating. Diluents are the most utilized additives in previous studies. However, reflective coatings may adversely affect the asphalt pavement in terms of strength; thus, a combination of materials is necessary to achieve adequate performance. Texture agents were added in polyester powder coating (PPC) and polyester/epoxy powder coating (PEPC) to modify their skid resistance. The inclusion of texture agents caused a negative effect on the cooling effect, adhesion strength, and abrasion resistance. Along with that, quartz sand, antifreeze, and levelling agents were added during the coating's manufacturing to enhance the pavement's skid and abrasion resistance in prolonging its service performance (JTTE Editorial Office et al., 2021).

There are various techniques that enhance the appearance of the heat-reflective coating to increase the coating's cooling efficiency on asphalt pavement. Chen et al. (2019) conducted an experimental study to investigate the cooling effects of a three-layer cool coating. The coating consisted of a reflecting layer, an emissive layer, and a thermal insulation layer. The researchers used self-made cooling effect assessment equipment for their study. Fig. 6 illustrates the conduction and convection of solar radiation on modified asphalt pavements with three-layer cooling coatings. The three-layer cool coating consists of (1) a reflective layer which reduces the light entering and passing through the coating while considering surface glare and reflects visible and infrared light, (2) an emissive layer, which serves primarily as a source of radiation, transfers energy to the surrounding area via thermal radiation, thereby lowering the temperature of the surface and mitigating the impact of the UHI, and (3) a thermal insulation layer that prevents heat transmission from the surface into the asphalt pavement. The primary functional materials were rutile titanium dioxide for reflecting, silicon dioxide for radiating, and hollow glass microbeads for insulating. Fig. 7 shows the optimum coating brushing quantities investigated by previous researchers with maximum and minimum quantities of 1.0 and 0.15 kg/m^2, respectively. The researchers from the selected articles in this study concluded that the ideal coating brushing quantity should range between 0.5 and 0.8 kg/m^2 to achieve adequate reflectance while being cost-effective.

3.2.1.2. Pavement materials modification. Pavement materials with cooling properties are characterized by lighter colours and more refined textures, contributing to a lower surface temperature. Research at Putrajaya, Malaysia was conducted to assess the efficacy of various paving materials, such as Blue Impala polished granite (BIPG) and Rose Tanggo polished granite (RTPG), in reducing the pavement temperature (Benrazavi et al., 2016). In contrast to BIPG, it was observed that the

Table 2
Data extraction results from reviewed articles.

Reference	Reflective material	Technology/method used	Property assessed	Max. temperature reduction	Other key findings
Santamouris (2013)	• White high reflective paints • Infrared colored paints • Heat reflecting paint • Color changing paints • Fly ash and slag	Heat-reflective paints were painted on the pavement surface or aggregates surface.	• Surface temperature reduction • Pavement albedo	Temperature reductions of 16 K–24 K (hot summer) & 2 K (night) by reflective layers using infrared reflective pigments.	• Increased the albedo of pavements/entire city.
Guntor et al. (2014)	• Wasted tiles (0.5–2.0 mm) • Sand • Epoxy resin	Epoxy resin layer and crushed tiles layer were evenly spread over the asphalt surface.	• Thermal performance • Surface reflectance & emissivity	Surface temperature reduction up to 4.4 °C.	N/A
Hu et al. (2015)	• Red, blue, and black thermochromic powders (3–10 mm)	Thermochromic powders were added to the original asphalt binder at 3%, 6%, and 10% by total weight of asphalt binder and mixed for 1 h.	• Conventional physical properties tests • Rotational viscosity • Dynamic shear test • Low-temperature cracking properties test	The utilization of thermochromic binders resulted in a decrease of 6 °C–8 °C in surface temperature.	• Thermochromic powders reduced penetration, phase angle, and creep rate & increased softening point, viscosity, complex modulus, rutting, fatigue, and stiffness of the asphalt binder. • High-temperature performance grades of asphalt binder.
Li et al. (2015)	• Slurry seal • Asphalt overlay • Chip seal • Cape seal • Sand seal	Reflective treatment replacements of 20%, 50%, and 100% by reflective material with an albedo of 0.25/0.3.	• Solar reflectance • Pavement albedo	N/A	• Typical pavement treatments have 0.05–0.30 steady state albedo. • Weathering and tracking with present technology will rapidly lower the albedo of treatments with reflective coating (0.7/higher).
Anting et al. (2015)	• Full body porcelain • Monoporosa/ceramic glaze • Porcelain glazes	The mixed-tile aggregates layer was applied after the epoxy layer coated on the pavement surface.	• Sunlight simulation • ANOVA analysis	The highest temperature difference was achieved by 100.0% full-body porcelain (4.9 °C).	• The optimal dosage of utilization has the potential to enhance the blending of waste tile aggregates to produce a cool-pavement coating substance.
Qin (2015)	• Light-colored aggregates • Chip seal • High near-infrared paint • White concrete components • Slag and fly ash • White topping	Resurfaced the asphalt pavements/concrete pavements using reflective materials.	• Surface temperature reduction	High near-infrared paint reduced surface temperature within 8 °C–20 °C at midday under hot weather.	• The reflectance index of colored pavements decreased by 20% after one year. • Thermochromic materials exhibit susceptibility to photodegradation when subjected open-air conditions.
Zheng et al. (2015)	Functional material: • Reflective (T), insulation (K), radiation (Y) Pigment and filler: • Inorganic black, organic black, inorganic red	Coating base paint was produced by blending resin (binder), functional materials and pigments. Anti-skid particles were spread on the base paint layer.	• Indoor cooling experiment • Outdoor cooling experiment • Anti-slide test	Indoor (Schemes 9 and 10) and outdoor (Scheme 5) cooling systems decrease the temperature by 30 °C and 10 °C, correspondingly.	• Scheme 5 is suitable for square and warning pavement applications; Schemes 7 and 8 are ideal for widespread use on asphalt pavement. • The addition of 1.18 mm machine-made sand achieved the best improvement on anti-skid performance of the coating applied.
Benrazavi et al. (2016)	• Blue Impala polished granite (BIPG) • Rosa Tanggo polished granite (RTPG) • Fontana concrete (FC) • Asphalt (AS)	Sites were chosen for a field study aimed at assessing how climate conditions, meteorological data, time periods, and landscape environments impact the surface temperature of pavement materials.	• Surface temperature measuring using infrared thermal imaging camera (FLIR E60)	Blue Impala polished granite (BIPG) was 15.5 °C cooler than asphalt (AS) during afternoon in open space (OS).	• BIPG and Rosa Tanggo polished granite (RTPG) surface temperatures increased in near water (NW) locations. • Granite with low porosity exhibit less water absorption and less heat transfer.
Rosso et al. (2016)	Five types of gravels origin from natural stone: • G1, G2, G3, G4 were light in color • G5 was slightly darker in color	The study examined how pedestrians perceive the selected reflective materials used for cool roofing and paving applications under varying summer weather conditions.	• Questionnaire (collecting data from pedestrians) • Surface temperatures • Thermal comfort • Visual comfort assessment	G4 with highest solar reflectance has the highest temperature difference (17.8 °C) with the asphalt surface.	• Light-colored gravels with high albedo (G1–G4) achieved higher thermal perception than the generic gravel G5. • Dark-colored gravels (G5) attained the highest visual perception of 1.6.
Rossi et al. (2016)	• White diffusive film (D1) • Beige diffusive film (D2) • Retro-reflective (RR) film	RR was initially assessed as a paving material in a small urban canyon, later confined within the canyon to analyse albedo distribution related to the heat island effect.	• Albedo value	N/A	• RR recorded an impressive 4.6% albedo boost, maintaining a consistent reflectivity pattern across the lid and showcasing superior near-perpendicular reflectivity.

(*continued on next page*)

Table 2 (*continued*)

Reference	Reflective material	Technology/method used	Property assessed	Max. temperature reduction	Other key findings
Anting et al. (2017)	• Full body porcelain (FBP) • Monoporosa (MP) • Porcelain glace (PG)	High-grade epoxy bound aggregate onto the asphalt, with surface temperature sensors monitoring pavement temperature.	• Solar reflectance • Thermal performance	Full body porcelain (M1) performed highest temperature reduction of 6.4 °C.	• M1 has the highest surface reflectance of 0.40 as it contains of brighter aggregate material. • High reflectivity provides low visual comfort to road users.
Cao et al. (2017)	• Fluorine monomers • Titanium dioxide (TiO_2)	An environmentally friendly fluorinated acrylate heat-reflective coating was prepared using conventional emulsion polymerisation method.	• Water absorption test • Gel rate test • Reflection spectrum test • Contact angle & FTIR test • Albedo analysis	The coated pavement achieved a temperature reduction of 13 °C.	• An optimal of 5% DFHMA demonstrated lower water absorption, thus reducing the risk of moisture damage to the pavement.
Gilbert et al. (2017)	• Slurry seal • Styrene acrylate reflective coating • Chip seal	Pavement life-cycle assessment (PLCA) tool was used to assess several case studies of city-wide pavement management practices over a 50-year life cycle.	• Air temperature reduction • Material changes • Building energy use changes	A 0.1 increase in urban albedo results in a summer peak temperature reduction of 0.11 °C, indicating a rate of 0.92 °C per 0.1 increase in albedo.	• The increase in pavement albedo resulted in an annual reduction in site energy consumption and cost savings through indirect cooling means.
Mohegh et al. (2017)	N/A	Weather research and forecasting (WRF) model version 3.5.1 was employed. Three scenarios (CTRL, COOL_LOW, COOL_HIGH) were simulated to study the influence of cool pavements on California climate.	• Changes in air temperature • Climate response • Pedestrian thermal comfort	The annual air temperature reduction for entire state of California was recorded as 0.18 °C–0.86 °C. Temperature reductions at 14:00 local standard time were found to be 0.32 °C.	• Albedo increased for 0.4, leading to reduction in annual air temperature. • The most significant reductions in temperature were observed during late morning and evening hours.
Sha et al. (2017)	• Film (polyvinyl alcohol, epoxy resin) • Pigment (titanium dioxide (TiO_2) with 10–20 vol%)	The solar heating reflective coating layer (SHRCL) was applied on surface of the asphalt pavement. The coating emulsions were exposed to ultraviolet rays for 12 h then observed by a fluorescence microscope.	• Cooling characteristics • Skid resistance • Ultraviolet radiation aging • Accelerated pavement testing	SHRCL lowered top temperatures by 10 °C ± 2.5 °C and bottom temperatures by 10 °C ± 3 °C.	• Sand enhances SHRCL skid resistance while retaining the cooling & lower abrasion resistance of asphalt pavement. • Persistent/inhomogeneous deformation cracks were found in certain SHRCLs.
Pisello (2017)	• Retro-reflective materials	Cool coatings on pavement surface provides passive cooling advantages.	• Solar reflectance • Heat capacity • Thermal conductivity	Surface temperature decreased up to 8.5 °C and air temperature reduction up to 7 °C.	• Solar reflectance was essential in affecting thermal comfort for mitigating UHI.
Kyriakodis and Santamouris (2018)	• Light-yellow thin layer asphalt (SR = 0.35 & emissivity = 0.9) • Elastomeric asphalt binder	Black conventional pavement was replaced by a thin yellow layer asphalt using slurry surfacing.	• Surface temperature	Cool materials had maximum 11.5 °C temperature reduction, especially during extremely hot conditions.	• Cool paving materials able to reduce ambient temperature maximum of 1.5 °C. • Cool asphalt reflectivity dropped drastically after six months.
Sen et al. (2019)	• New asphalt concrete • Reflective concrete • Reflective coating	Five types of surfaces were assessed in terms of their albedo and temperature reduction.	• Surface temperature • Albedo analysis	Outdoor environment with reflective coatings achieved substantial temperature reduction of 0.8 °C–1.0 °C.	• A reflective coating can increase the albedo values up to 0.60.
Xie et al. (2019)	• Titanium oxide (TiO_2) • Red iron oxide • Organosilicon-modified acrylate emulsion	Heat-reflective coating with near-infrared fillers and pigments was applied onto the porous asphalt samples.	• Physical properties • Spectral reflectance • Solar reflectance • Skid & abrasion resistance	The optimum incorporation achieved surface temperature reduction of 9 °C compared to control sample.	• The ideal coating formula is B(red iron oxide): C(R-TiO_2):A(NIR-TiO_2) = 5:25:8. • Coated pavements are smoother and thus reducing skid-resistance and anti-abrasion.
Yi et al. (2019)	• Epoxy resin (heat-reflective coating substrates) • Modified amine agents • Binder (polyurethane modified epoxy resins, high-elasticity epoxy resins) • Fillers (pigments, reflective filler)	1. Materials were analysed and mixed to form road heat-reflective coating (RHRC). 2. Adhesion promoter and wear-resisting agent were added to produce super RHRC (SRHRC). 3. SRHRC was assessed via experimental tests.	• Indoor & outdoor temperature test • Bond strength • Abrasion resistance • Skid resistance	White coatings exhibit superior cooling properties by achieving temperature reduction of 11.7 °C.	• White and yellow coatings exhibited the best cooling effect but lowest visual comfort for road users. Red coating is more recommended. • The coating lowered road surface anti-skid performance, yet it met traffic safety standards.
You et al. (2019)	• Nanometallic materials (CuO nanoparticles-Cuo NPs) are the selected pigments • Three shapes of Cuo NPs (spherical, rod, &flake) • Reflective coatings (E51 epoxy resin, polyamide hardener)	There are 12 types of black coating formulated using three types of CuO NPs and dosages at 6.8%, 12.8%, 18.0%, and 22.6%.	• Surface temperature differences	Flake CuO NPs with mass percentage of 22.6% reduced the surface temperature of asphalts by a maximum of 12.62 K and 10.04 K in typical summer and autumn meteorological condition, respectively.	• Near-infrared radiation (NIR) solar reflectance rises with CuO NP mass percentage, although it eventually declines. • Economically, 22.6 wt% is optimal.

(*continued on next page*)

Table 2 (*continued*)

Reference	Reflective material	Technology/method used	Property assessed	Max. temperature reduction	Other key findings
Chen et al. (2019)	• Rutile titanium dioxide • Silicon dioxide • Hollow glass microbeads • Epoxy resin	Titanium dioxide was incorporated at 4.5%, 9.0%, 13.5%, and 18%. The coating comprises three distinct structural layers. The initial layer comprises the reflective layer, followed by the emissive layer, and ultimately, the thermal insulation layer.	• Surface temperature determined using self-developed cooling effect evaluation device (CEED)	The reflective layer lowers the sample surface temperature by 8.18 °C. An emissive layer achieved temperature reduction of 11.1 °C. The bottom thermal insulation layer exhibits a cooling effect of up to 13.25 °C.	• The cooling effect of the reflective material was superior regardless of the asphalt mixture's depth. • The cooling effect demonstrates an initial upward trend followed by a gradual decrease as the depth increases.
Jiang et al. (2020)	• Base polymer • Additives (diluent, wetting dispersant, defoamer & flow agent) • Functional materials (high-refractive-index rutile titanium dioxide powder) • Colouring pigments (iron oxide red = 3.5%)	Heat reflective coating (HRC) and thermally resistant and heat-reflective coating integrated pavement (TR-HRC) samples was prepared by painting the reflective layer to their surfaces.	• Thermal resistance test • Rutting test • Skid-resistance test • Accelerated abrasion test	TRFC-13, HRC, and TR-HRC samples were observed to decrease the indoor temperature by 6.8 °C, 11.2 °C, and 13.2 °C, respectively.	• TRFC-13, HRC, and TR-HRC specimens exhibited not only lower road surface temperatures but also improved dynamic stability compared to ordinary SMA-13 samples. • The inclusion of machine-made sand, ranging in sizes from 0.6 mm to 1.18 mm at 40 wt%, maximally enhanced pavement skid resistance.
Zheng et al. (2020)	Matrix material: • Matrix material (E-51 epoxy resin, 593 curing agent) • functional fillers (inorganic pigments, titanium dioxide, floating beads) • Admixture (diluent, defoamer, dispersant)	Several dosages combinations (10%–15% TiO_2, 10%–30% floating beads, 30% pigment, 0.6% dispersant, 0.3% defoamer, 25% curing agent, and 10%–35% diluent) were produced and applied to the pavement surface.	• Reflectivity test • Cooling performance • Coating abrasion test • Pavement service performance test	15% titanium dioxide and 10% floating beads reached a maximum temperature difference of 9.7 °C, 8.3 °C and 6.6 °C at the upper surface, 2 cm depth and the bottom of specimen, respectively.	• Green coating had the highest reflectivity and cooling impact, followed by red, grey, and blue. • The addition of floating beds enhanced the wear resistance of the heat-reflective coating. • An appropriate coating dosage should be regulated at 0.4–0.8 kg/m^2 to ensure safe driving.
Li et al. (2020)	• TAP (a silicate mineral consists of mainly Al_2O_3, SiO_2, B_2O_3, Fe_2O_3, MgO, FeO, Na_2O)	Tourmaline anion powder (TAP) was used as cooling modifier for asphalt due to its negative ion release, thermoelectric, and spontaneous polarization properties.	• Dynamic shear rheometer • Bending beam rheometer • Temperature difference test • Dynamic modulus • Semicircular splitting • Water stability test	The indoor and outdoor test showed a maximum cooling value of 6.6 °C and 7.2 °C, respectively on the modified pavement.	• TAP-modified asphalt's high-temperature stability rises. • The low-temperature fracture resistance of modified pavement varies and then falls. • The optimum TAP content was determined to be 16%.
Middel et al. (2020)	• Solar reflective coating	Thermal performance of solar reflective pavement was monitored hourly between 11:30 LST and 20:00 LST at several sites using the mobile instrument platform MaRTy.	• Hourly radiation flux observations	The reflective coating effectively lowered the road surface temperature at all observed times, resulting in a 6.0 °C reduction during midday, a 4.0 °C decrease in the early evening, and a 1.6 °C–1.8 °C drop after sunset.	• On average, coated asphalt concrete exhibited a shortwave radiation reflection of 118 Wm^{-2}, which increased to a maximum of 168 Wm^{-2} during midday.
Huang et al. (2020)	• Ceramic waste (derived from waste ceramic tiles from the demolition of buildings) • Lignin fiber • Limestone filler	Ceramic waste aggregate with low thermal conductivity was used at 10%, 20%, 30%, 40%, and 50% by volume as a substitute for coarse aggregates in stone mastic asphalt (SMA).	• Wheel rutting test • Freeze-thaw indirect tensile test • Bending beam test • Thermophysical parameter test (hot disc method test) • Anti-skid performance test • Fatigue beam test • Thermal insulation test	Modified pavements (CASMA-13) had higher surface temperature (68.9 °C–70.9 °C), but lower temperature at interface (58.7 °C–63.1 °C) and bottom (42.5 °C–46.9 °C) specimen than the control samples (SMA-13) (68.5 °C, 66.1 °C, and 51.6 °C, respectively).	• CASMAs outperformed SMA in interior thermal insulation by showing higher temperature difference between surface/interface and surface/bottom. • The thermal insulating layer of CASMAs demonstrated a lower degree of permanent rutting deformation when subjected to repeated traffic loads.
Roxon et al. (2020)	N/A	A bottom-up regression statistical approach was employed to conduct a 12-year financial and environmental analysis of the urban heat island (UHI) effect across 48 US states.	• Energy consumption • Carbon dioxide emission • Energy expenditure	N/A	• For the total 48 US states, the expenditure and carbon emissions due to the UHI effect were $120B and 550 M tons each year, respectively.
Du et al. (2020)	• Hollow glass microsphere (HGM) • Limestone mineral filler (LMF)	Asphalt mastics with inclusion of HGM (replacing LMF) was yield at a filler-to-asphalt ratio of 1.2.	• FTIR, XRF, SEM tests • Thermal conductivity test • Spectral reflectance test • Multiple stress creep recovery (MSCR) test • Linear amplitude sweep (LAS) test	N/A	• HGM improves asphalt mastic heat resistance by achieving a lower thermal conductivity (−40%) than the conventional mastic. • HGM increases asphalt mastic's near-infrared reflectivity by 60% higher than the conventional mastic.

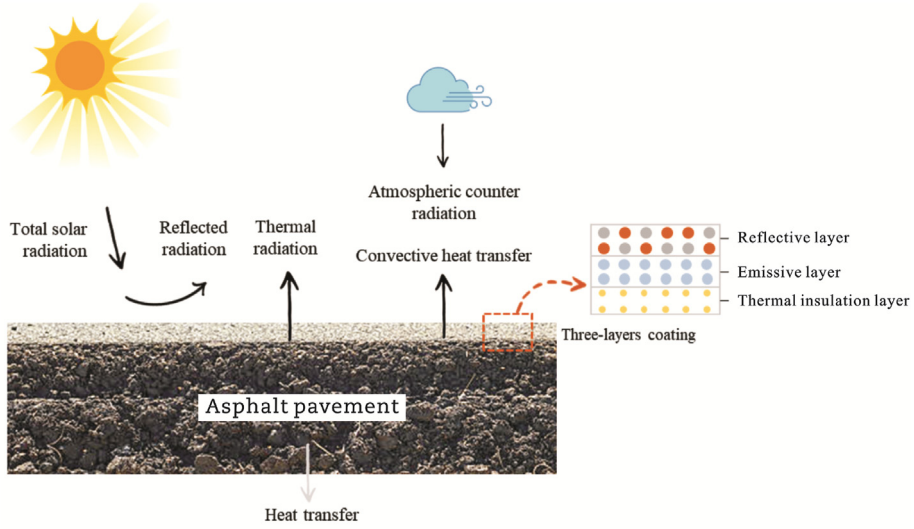


Fig. 6. Structure and principles of three-layer cool coating on asphalt pavement (Chen et al., 2019).

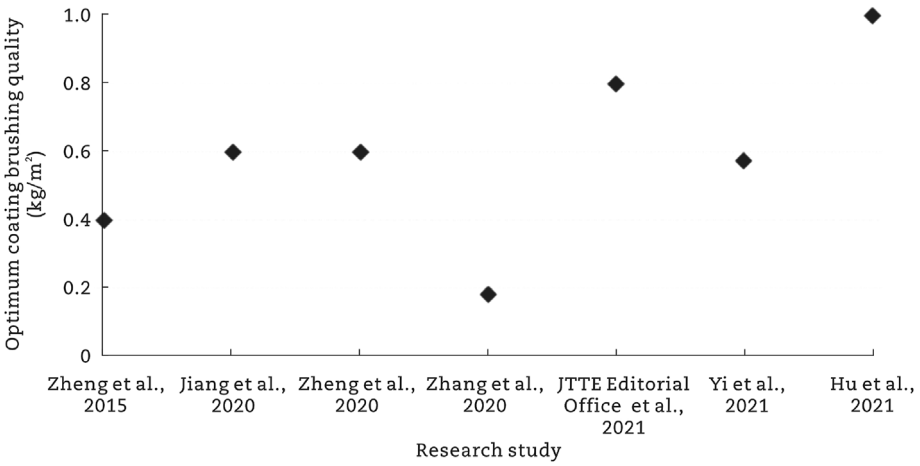


Fig. 7. Optimum coating brushing quantity among the researchers.

surface temperature of ordinary asphalt, which exhibited a temperature increase of 15.5 °C, demonstrated the highest temperature among other pavement materials. In addition, Alghamdy et al. (2021) presented a thermophysical evaluation of the unique white dolomitic marble, also known as Thassos Marble, used for pavement in Makkah City, Saudi Arabia. Fig. 8 compares the cooling performance between the different paving materials (Thassos Marble, concrete, and asphalt). It is evident that the maximum temperature attained by the marble was comparatively lower than the ambient temperature of the surrounding atmosphere. These studies showed that pavement materials with cold, polished surfaces or the perceived whiteness of the paving materials might retain a high degree of solar reflectance, significantly decreasing midday solar heating. Besides, these paving materials also have a high heat conductivity, resulting in the efficient dissipation of the previously absorbed solar insolation.

In addition to the cool-coloured gravel used in the pavement to produce a high albedo value, the field albedo will be affected by the grain size of the gravel. Table 3 displays the cooling properties of various gravel particle sizes. It was proved that smaller-grained gravel had a high albedo level as well as a high amount of solar reflection. Nevertheless, Rosso et al. (2016) discovered a modest glare risk associated with high levels of gravel albedo regarding visual perception, and it was determined that individuals prefer grassland. Among the samples, G5 exhibited the lowest albedo but the highest perceived rating, with a score

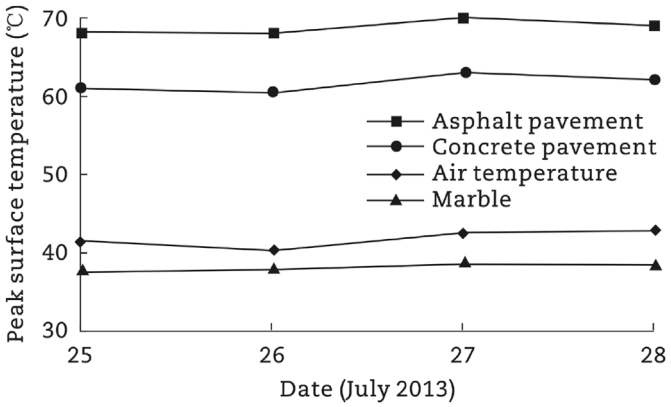


Fig. 8. Peak surface temperature (afternoon time) for different paving materials (Alghamdy et al., 2021).

of 1.6. The other samples, namely G1, G2, G3, and G4 had perceived ratings of 2.2, 2.9, 2.9, and 4.1, respectively. The study revealed a negative correlation between albedo and visual perception, whereby higher albedo values were associated with decreased visual perception. Notably, the surface denoted as G4 exhibited the poorest visual

Table 3
Composition and characterization of gravel (Rosso et al., 2016).

Gravel type	Name	Grain size (mm)	Solar reflectance (%)	Albedo (%)
Local natural stone with high reflectance	G1	8.0–22.4	38	36
	G2	4.0–12.5	50	39
	G3	2.0–5.6	45	40
	G4	0.0–4.0	62	44
Common mixed natural gravel	G5	4.0–12.5	27	29

perception. The significance of visual perception must be considered as it may have a major impact on the degree of outdoor comfort in highly urbanized environments, particularly in warm climates.

Utilization of waste materials has substantial economic and environmental advantages over incorporating more prevalent resources. The utilization of waste tile aggregates to develop cool-pavement coating materials aimed at reducing pavement surface temperature was explored by Guntor et al. (2014) and Anting et al. (2015). In addition, the feasibility of replacing natural aggregate with ceramic waste in asphalt mixtures was investigated by Huang et al. (2020). The impact of ceramic waste-based asphalt mixture's thermal insulation properties on addressing rutting defects and mitigating the effect of heat islands was focused on their research. It was revealed that the cooling asphalt pavement's permanent rutting deformation was 45.36% lower than conventional pavement, Furthermore, ceramic waste aggregate exhibited exceptional cooling capacity. Using ceramic waste aggregate as a thermal insulation layer in asphalt pavement experienced superior performance in preventing rutting and cooling. This considerably increases the competitiveness of the asphalt pavement option that reflects heat. The application of waste ceramic tiles to replace the natural aggregate in stone mastic asphalt achieved a significant reduction of 8.8 °C (Huang et al., 2020) compared to the other options, which were only 4.4 °C (Guntor et al., 2014) and 4.9 °C (Anting et al., 2015), respectively.

3.2.1.3. Pavement treatment using light colour material. When pavement is sealed with light-coloured materials, the surface temperature is reduced, and solar reflectance is enhanced. During chip sealing process, a thin layer of binder is applied onto the deteriorated pavement surface, followed by the construction of finely graded aggregate as the foundation. Over time, the albedo of a chip seal surface undergoes a reduction, however, it stays higher than standard asphalt pavement for approximately five years. Implementing this approach can decrease the temperature of asphalt pavements by as much as 9 °C (Li et al., 2015).

Li et al. (2015) assessed various pavement treatments used in California. These treatments included slurry seal, sand seal, chip seal, cape sea, asphalt overlay, and asphalt restoration, each with constant steady state albedo value of 0.09, 0.10, 0.15, 0.06, 0.10, and 0.10, respectively. Among the treatments, slurry seal is the most common treatment utilized by the local government with 28%–82% of mileage treated. It is found that chip seal performed the highest albedo of 0.15 among the six main treatments (slurry seal, sand seal, chip seal, cape seal, asphalt overlay, and reconstruction), thus it is feasible to modify the chip sealing treatment using high-albedo materials to improve its cooling capacity.

Anting et al. (2017) conducted a study focused on the development of a heat-reflective pavement, utilizing three specific waste tile aggregates, namely full body porcelain (FBP), Monoporosa (MP), and porcelain glaze (PG). These tile aggregates were chosen based on their distinct surface colours, contributing to the variability in their reflective properties. The researchers explored five different formulations of these selected tile aggregates. A high-grade epoxy was employed as the binding material to effectively secure the aggregate materials onto the asphalt surface.

3.2.1.4. Asphalt binder modification. Modified asphalt binders with incorporated cool materials may increase the cooling ability of asphalt pavements. Thermochromic materials are substances that undergo reversible colour changes in response to variations in temperature. In hot summers, it may reduce the pavement's surface temperature, while in cold winters, it can increase the temperature. Thermochromic materials were fabricated using organic mixtures comprising lueco dye, which acts as an electron donor, developer (an electron acceptor), and solvent. The interaction between the developer and leuco dye below the transition temperature, causing a shift in the absorption peaks from the ultraviolet to the visible spectrum, leading to the pigmentation of the powder. At temperatures beyond the transition temperature, where solvent-developer interactions dominate, the leuco dye undergoes separation from the developer, and experiences a loss of colour. The encapsulation of thermochromic materials by trioctanoinand leads to a phenomenon where thermochromic powders exhibit a white or light-coloured appearance when exposed to high temperatures. Implementing thermochromic binders can enhance the functionality of asphalt pavement by mitigating the surface temperature, particularly in cities with hot climatic zones. Based on the comprehensive analyses of binder performance, cost, and constructability, it is recommended that the use of thermochromic powders be limited within the range of 5%–6% (Hu et al., 2015).

In addition, utilizing tourmaline anion powder (TAP)as a modifier can generate a modified binder that exhibits surface cooling properties. TAP is a silicate mineral comprised of Al_2O_3 (35.1%), SiO_2 (34.8%), B_2O_3 (11.0%), Fe_2O_3 (10.2%), MgO (4.7%), FeO (1.4%) and Na_2O (0.9). It exhibits piezoelectric, thermoelectric, and negative ion-releasing properties. The utilization of TAP-modified binder has been observed to reduce the temperature of asphalt pavement during hot weather conditions, thereby preventing the occurrence of rutting. Additionally, this binder has been found to enhance the performance of asphalt at high temperatures. The TAP-modified binder exhibits a maximum surface cooling value of 5.9 °C, further enhanced to 7.2 °C during outdoor testing. The optimum incorporation of TAP is 16% of the asphalt binder.

Apart from that, incorporating an infrared-reflecting pigments into an elastomeric asphalt binder produces a yellow slurry film with a thin consistency, which is applied as a coating on the asphalt. During the summer, cool materials were discovered to have high-temperature reduction until 7.5 °C. According to Kyriakodis and Santamouris (2018), it has been discovered that using cool paving materials can lower the emission of sensible heat into the environment by a maximum of 1.5 °C. In the research of Du et al. (2020), the limestone mineral filler in the asphalt mixture was replaced with hollow glass micro-spheres (HGM). HGM is a viable asphalt pavement cooling agent, and its incorporation has been shown to enhance the fatigue durability of asphalt mastic.

3.2.2. Mechanisms and impacts of heat-reflective asphalt pavements

3.2.2.1. Mechanisms of heat-reflective properties. Continuous development of hard, heat-absorbent surfaces, urbanisation, and reducing natural vegetation are the primary sources of the UHI impact (Sen et al., 2019). Fig. 9 demonstrates the formation of UHI due to the atmospheric heat, advanced technology and high surface temperature of asphalt pavements. As a result, cool pavements are anticipated to lower asphalt pavement temperatures to counteract the UHI impact. Several critical factors, including thermal conductivity, heat capacity, solar reflectance, emissivity, shading, and evaporation rate, primarily influence the thermal performance of cool pavements. These characteristics may influence the temperature of subsurface pavement, the temperature of the near-surface air, the reflection and emission of radiation, thermal comfort, as well as the quality of the air at the surface (Li et al., 2014).

The implementation of heat-reflecting asphalt, coupled with a thermal insulation coating, has the potential to decrease the temperature of pavement and mitigate the UHI effect. This is achieved through the enhancement of sun reflection and the facilitation of air filtration. Asphalt pavement with a heat-reflecting coating has a high capacity to reflect visible and infrared solar radiation energy. For instance, pigments

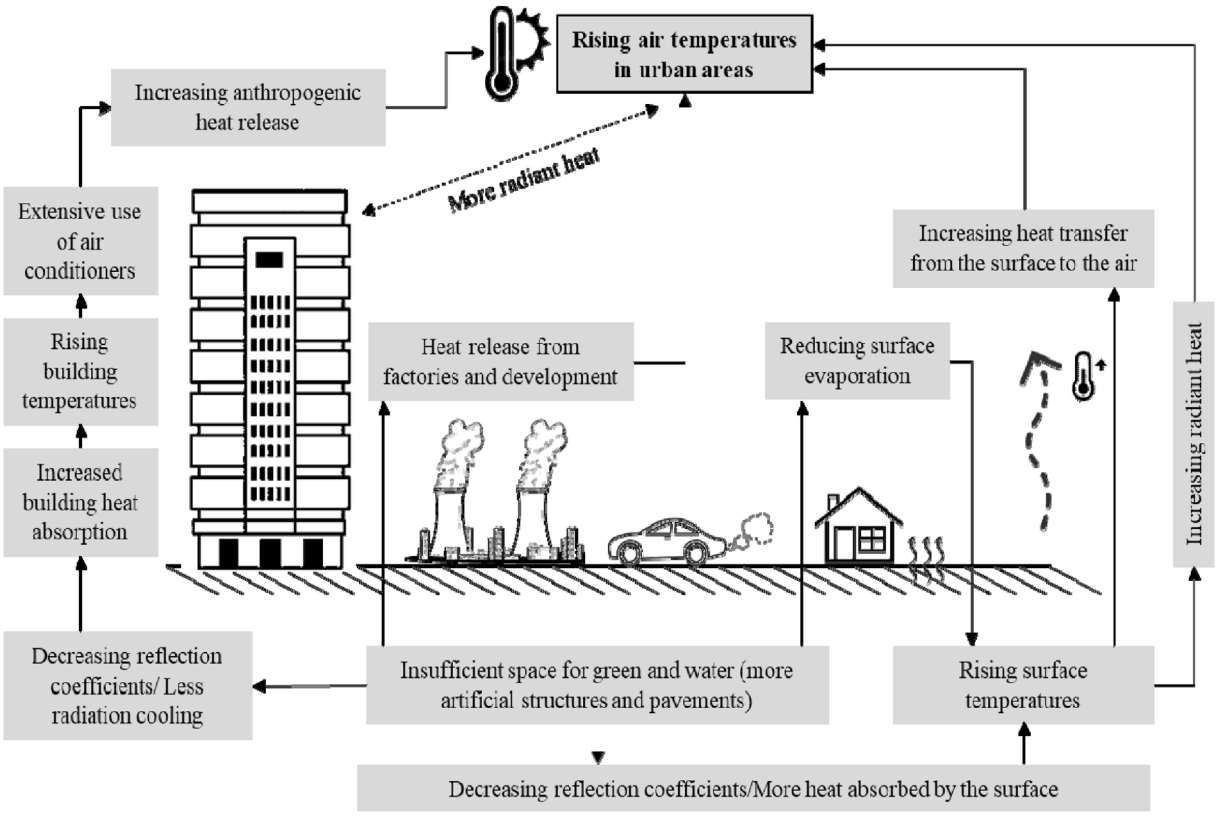


Fig. 9. Formation of UHI effect due to urban development (Mohajerani et al., 2017).

in the coating reflect light well in both the visible and near-infrared spectrums (0.4–0.76 μm and 0.76–2.5 μm, respectively). The pigments can produce long-wave radiation between 2.5 and 15 μm that reflects the absorbed heat, resulting in a remarkable temperature drop without the need for energy. Fig. 10 depicts the mechanisms of the solar reflective coating in reflecting the solar radiation energy that consists of two different regions: visible region and infrared region. Table 4 presents the proportion of radiation reflected. From the results, the application of heat-reflective coating onto asphalt pavement has demonstrated efficacy in reflecting a significant proportion of incident solar radiation.

Besides, large void-coated pavement can potentially function as cool pavement to alter the pavement surface temperature and heat transfer capabilities. Generally, a cooling pavement comprises a heat-reflective coating layer and a void-rich asphalt pavement, as depicted in Fig. 11. The significant voids present in the asphalt pavement served to reduce the amount of heat that was conducted into the road surface. This, in turn, promoted the convection of air and road heat, thereby enhancing

Table 4
Wavelength interval and proportion of solar radiation (Chen et al., 2022).

Light region	Visible region	Infrared	Ultraviolet region
Wavelength interval (μm)	0.40–0.76	0.76–2.50	<0.40
Proportion of radiation energy (%)	50	43	7

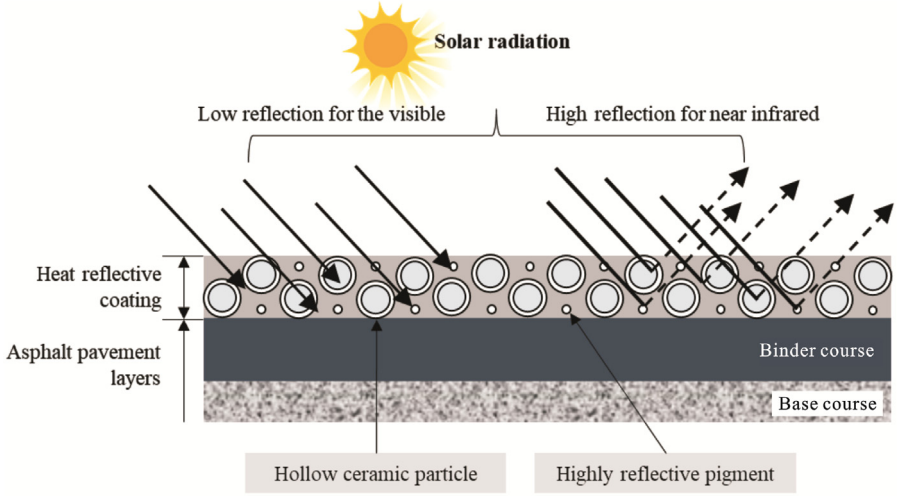


Fig. 10. Mechanism of solar reflective coating (Zheng et al., 2015).

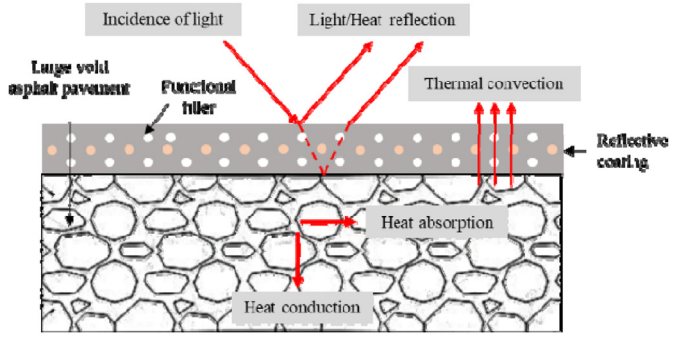


Fig. 11. Cooling mechanism of large void asphalt pavement (Zheng et al., 2020).

the road surface's capacity to dissipate heat. This application can effectively lower the surface and interior temperatures of the pavement.

Asphalt binders modified with thermochromic materials will also respond differently to solar radiation compared to conventional traditional asphalt binders. Ultraviolet, visible, and infrared light rays are among the range of wavelengths that produce solar energy, which accounts for over fifty percent of the total amount. The modified asphalt binder should exhibit decreased solar reflectance (and thus increased solar absorption) below the transition temperature of thermochromic powder, remarkably when the powders are less infrared-reflective, as illustrated in Fig. 12(a). Meanwhile, the binder should be reflected more to infrared at a higher transition temperature, particularly when the particles exhibited heightened reflectivity to infrared, as demonstrated in Fig. 12(b). The thermochromic asphalt binder possess energy-reflective and energy-absorbing properties that aid in maintaining the pavement's durability and performance by keeping it within the required optimum temperature range.

3.2.2.2. Solar reflectance effects on asphalt pavement. According to Sen et al. (2019), the application of a reflective coating has the potential to substantially elevate albedo values, with an increase observed up to 0.60. Albedo refers to the reflective properties of a surface, specifically its ability to reflect sunlight. The albedo of the asphalt pavement plays a crucial role in determining the amount of solar energy the surface reflects. The higher the albedo value, the more sunlight a pavement reflects. Reflective quality is particularly important in heat-reflective asphalt pavement, as it helps to lower the pavement surface temperature, preventing the pavement from absorbing and retaining excessive heat (Jiang et al., 2020). Therefore, lowering the surface temperature by increasing the albedo may be a good idea, particularly in climates and seasons characterised by elevated solar radiation levels. Based on existing studies, albedo value close to 0.1 and 0.5 will probably have surface temperatures close to 71 °C and 53 °C, respectively (Santamouris, 2013). A conventional pavement is commonly made of dark colour asphalt, presents relatively low steady state albedo, ranging from 0.04 to 0.06 (Li et al., 2013, 2015; Anting et al., 2017). It absorbs huge amount of heat emitted from the solar radiation during the daytime and reradiate the heat during the nocturnal period. Therefore, asphalt pavements usually experience a high surface temperature during the daytime, ranging from 48 °C to 67 °C.

The solar reflectivity and albedo of cool materials is often influenced by various physical properties such as sizes and colours of the particles. Cool materials, characterized by white or light-coloured surfaces, exhibit high albedo in the visible spectrum compared to darker surfaces. Rosso et al. (2016) established a correlation between solar reflectance and particle size. Gravels with light surface colour originated from an almost white cave and a grain size ranging from 0 to 4 mm demonstrated higher solar reflectance (62%) and albedo values (44%) as determined by spectrophotometer and albedometer, respectively, in comparison to the non-white gravels with sizes ranging from 8.0 to 22.4 mm, which exhibited lower solar reflectance (38%) and albedo (36%). At the same time, waste tile aggregates with a brighter surface demonstrated increased solar reflectance in both the visible light spectrum and near-infrared reflectance. This characteristic serves to reduce heat penetration into the pavement, attributed to the material's heightened reflectance in the near-infrared region (Anting et al., 2017).

Furthermore, a titanium concrete pavement sample, characterized by a distinct clear grey colour with a subtle white undertone and constructed using White Portland cement, exhibited the highest albedo (0.57) among the other samples (Sanjuán et al., 2022). Notably, the concrete pavements with light grey colour displayed significantly higher albedo compared to an asphalt pavement with a darker black colour. The solar reflectance index (SRI), a key indicator of a material's cooling efficiency, displayed a linear correlation with its albedo value. The titanium precast concrete pavement consistently achieved the highest SRI across different wind speeds. However, it's important to note that over time, the albedo value experienced a decline due to weathering and the accumulation of dirt on the surface (Li et al., 2015; Sanjuán et al., 2022). This observation underscores the dynamic nature of surface properties and the need for ongoing maintenance to preserve the reflective characteristics of cool materials over extended periods.

An increased albedo by 0.1 is estimated to reduce the maximum pavement surface temperature by more than 3.0 °C. Asphalt pavements featuring a bright surface, such as those coated with white topping (commonly composed of Portland cement concrete that often comprises of fibres for better performance), exhibit a notable increase in albedo, approximately 0.40 higher than that of conventionally freshly paved asphalt (Croce et al., 2021). Generally, the initial albedo of reflective coating can reach about 0.7 or higher, however, the albedo value will decrease rapidly to a low steady state value due to weathering (Li et al., 2015).

3.2.2.3. Pavement performances of heat-reflective pavements. The reflectivity characteristics of the heat-reflective pavements not only improve the thermal characteristics of asphalt pavements, but also contribute to enhance the overall pavement performances. The ability of heat-reflective properties to divert solar radiation rather than absorb it minimizes the thermal stresses endured by the asphalt, especially during the daytime (Seifeddine et al., 2023). This reduction in thermal stress contributes significantly to mitigating wear and tear, thereby enhancing the overall durability of the pavement, and ensuring superior long-term performance.

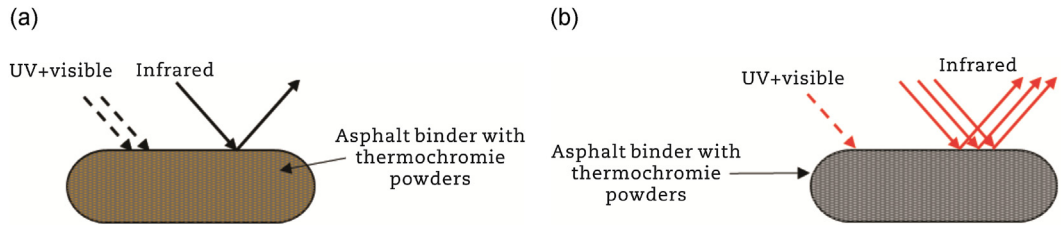


Fig. 12. Schematic illustration of the reflected radiation mechanism of asphalt binder containing thermochromic powders (Zhang et al., 2021). (a) Lower infrared reflectivity under transition temperature. (b) Higher infrared reflectivity above transition temperature.

Epoxy polymer is often used as the adhesive component in heat-reflecting coatings. This contributed to its chemically inert properties, high fatigue strength, excellent transparency, and good infrared shielding performance as long as the connected surface is cleaned meticulously during the implementation (Chen et al., 2022; Sha et al., 2017). Moreover, acrylic ester polymers demonstrate excellent plasticity and UV stability, which is advantageous for asphalt pavement (Pukánszky, 2016). In addition, the thermochromic powders improved the solar reflectance ability of asphalt binders. According to Hu et al. (2015), the thermochromic asphalt binders acquired a higher binder stiffness at room temperature, resulting in enhanced softening point and viscosity values. This suggests that the powders strengthen the asphalt binder's resistance to plastic deformation, rutting, and fatigue. The inclusion of less than 5%–6% thermochromic powder improves the performance of thermochromic asphalt binders at high temperatures (Hu et al., 2015).

Furthermore, cool materials such as hollow glass microsphere (HGM) and coarse ceramic waste aggregate (CASMAs) are useful in improving the asphalt pavement performance. The fatigue resistance of HGM asphalt mastic was greatly increased with the escalation of strain level from 2.5% to 5.0%. The mixing procedure resulted in the fracturing of the hollow centred HGM particles, which caused a reduction in the required volume of HGM compared to the anticipated volume. Hence, the HGM-based asphalt mastic included a greater amount of asphalt, enhancing fatigue resistance (Du et al., 2020). At the same time, the asphalt pavement with CASMAs thermal insulation coating exhibited superior resistance to rutting while simultaneously providing cooling benefits, according to Huang et al. (2020). When subjected to repeated traffic loads, the cooling asphalt pavement exhibits a permanent rutting deformation of 45.36% lower compared to conventional pavement.

In addition, the utilization methods of the reflective coating influenced the performance of pavement. The thermosetting powder coating's (TPC) cooling effect and anti-glare properties are comparable to those of the conventional heat-reflective coating, while its mixture performance, as measured by skid resistance, abrasion resistance, and adhesion strength, exhibits superior characteristics. TPC was sprayed onto the surface of asphalt mixtures at higher temperatures, thus achieving a stronger bond between the coating and asphalt mixtures. The superior performance and cost-effectiveness allow the TPC to be suited for constructing reflective pavement (Zhang et al., 2020).

Other than that, heat-reflective pavement with coated hot-pressed hydrated lime (HL) also performed well in extreme conditions. The performance of the heat-reflective pavement regarding the skid, abrasion, and erosion resistances was significantly influenced by the spreading dosages, particle sizes, and organic modifications of HL. Laboratory tests demonstrated the reduction in particle sizes of HL have increased the erosion resistance of the asphalt mixtures. The optimal incorporation of HL with a spreading dosage of 100 g/m^2, a particle size of 1200 mesh, and coated via graft polymerisation modification (GPM) resulted in higher performance in both abrasion resistance and erosion resistance for 50% and 55%, respectively (Man et al., 2021). Functional fillers such as silica, diatomite, and hollow glass beads, were used as a substitute for the costly titanium dioxide in production of a heat-reflective coating. According to the findings, it can be inferred that the reflective coating has high pavement performance concerning skid-resistance, resistant to abrasion, and water permeability (Chen et al., 2022). In short, using heat-reflective pavement is not just advantageous in minimizing the UHI effect. Yet, it may also enhance the pavement strength regardless of the application method.

3.2.2.4. Drawbacks. Due to weathering and traffic, the albedo of pavement tends to vary over time, particularly in the first month following installation (Du et al., 2018; Li et al., 2015). The high initial albedo of pavements with the reflective coating will swiftly and considerably decline to a low value owing to weathering and traffic loads (Li et al., 2015). During a laboratory assessment, the cooling pavement with four different coating colours was subjected to 10,000, 20,000, and 30,000 times of wear, respectively. The cooling performance tended to have deteriorated with the increase in wear times, but it remained steady after 20,000 wear cycles. To preserve the heat-reflective coating's capacity to reflect solar radiation, it must be routinely maintained. However, according to Sha et al. (2017), persistent or inhomogeneous deformation may cause the reflective coating layer to sustain early deterioration, including cracking and spalling. The primary reason for the differential in stiffness or the modulus is the variation between asphalt binders and resins.

In fact, the coating will reduce the road's potential to resist slippage (Hu et al., 2021; Zhang et al., 2021). Increasing the coating dosage reduced the pavement's skid resistance and seepage coefficient (Zheng et al., 2020). Nevertheless, essential measures are required to counter the issues. Hu et al. (2021) demonstrated that the dispersion of silicon carbide particles can considerably enhance the anti-slip properties of the coated surface. The ideal allocation rate was determined to be 1 kg/m^2, where the pavement's anti-slip coefficient significantly surpassed the minimum threshold required for safe asphalt driving. In addition, the use of machine-made sand-sized 1.18 mm with a spreading dosage of 0.16 kg/m^2 may also boost the coating's skid resistance without diminishing its cooling capacity (Sha et al., 2017; Zheng et al., 2015).

The reflection of radiation by these pavements may significantly impact human health. The human eye' corona is noticeably altered by the reflection of the radiation in the visible spectrum, adversely affecting drivers' perception. Meanwhile, the reflected UV rays may pose a potential hazard to the human skin (Georgakis et al., 2014). Aside from that, Anupam et al. (2021) and Cheela et al. (2021) commented on glaring issues due to the high reflectance from the reflective coating have been the most common issues found with reflective pavement. According to a survey of pedestrians, the use of gravel with less reflectivity is favoured, with more than 80% of respondents considering it pleasant and bearable. Yet, the gravel with the maximum reflectivity is the least desirable (Rosso et al., 2016). Therefore, road users should be cautious to prevent an unpleasant visual experience while traversing gravel pavements under bright weather.

However, reflective pavements are typically cooler than regular pavements under limited conditions. The degree of radiation reflection directly impacts the extent of temperature reduction. Fig. 13 shows that the reflective pavements were only effective if the incident radiations were reflected in the atmosphere. Alternatively, the radiation's reflection can negatively impact the thermal comfort of nearby buildings, cars, and pedestrians (Anupam et al., 2021). Construction and outdoor workers exposed to the reflective pavement during peak solar hours may experience adverse effects (Anand and Sailor, 2022). A vegetated setback is necessary to separate the roadways and sidewalks to mitigate the high heat load on sidewalks in the early evening. Despite the reflective pavement is environmental-friendly, the municipal goals of promoting

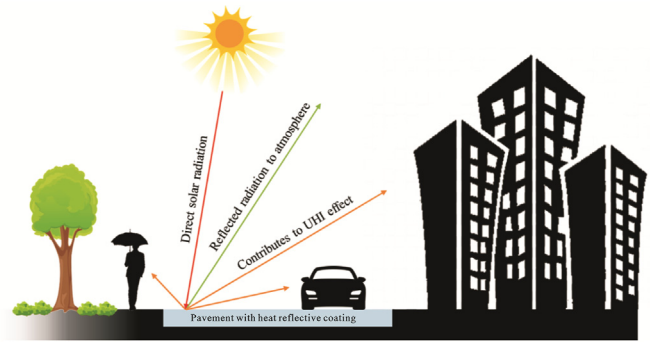


Fig. 13. Heat reflectance mechanisms of the existing heat-reflective pavement (Anupam et al., 2021).

walkability and outdoor activities to improve residents' health are not advanced using the cool pavement. Therefore, the current cool reflective pavement technologies require modifications to fit the surrounding restrictions.

3.2.3. Performance assessment in UHI mitigation with heat-reflective asphalt pavements

3.2.3.1. Pavement surface temperature. Typically, pavement surface temperature is known to have a significant influence on the long-term efficacy of paved surfaces. Researchers had reviewed a number of articles analysing the efficacy of each method of application for reducing UHI to select the most appropriate heat-reflective asphalt pavement for roads. During straightforward performance analysis, the cooling performance of this technology is the most important factor to consider, as this parameter demonstrates the system's efficacy. The cooling effectiveness, or the reduction in surface temperature, of heat-reflective pavement varies based on its method of application and technique of installation.

From the analysis of all 43 articles in this SLR, it is evident that the application of heat-reflective coatings emerges as the most viable strategy for mitigating pavement temperatures. More than half of the reviewed articles advocate for this approach. Reflective coatings that usually have increased albedo, effectively reduce the absorption of solar radiation, resulting in lower surface temperatures (Schneider et al., 2023). Nevertheless, it is crucial to note that while these reflective coatings demonstrate effectiveness in temperature reduction, they often present challenges related to glare, thereby compromising human visual comfort. Despite their efficacy in managing pavement temperatures, the potential discomfort caused by glare must be carefully considered in the implementation of heat-reflective coatings.

As the albedo of a surface increases, there is a corresponding decrease in the maximum temperature reached by the pavement surface. This relationship was highlighted by Qin (2015), who specifically noted a linear correlation between the absorptivity of the pavement and the maximum temperature attained by its surface. In other words, when the pavement's ability to reflect sunlight and absorb less heat (higher albedo) is enhanced, the peak temperature experienced by the pavement surface decreases in a predictable and linear fashion.

Heat-reflective coatings such as retro-reflective (RR) materials (Pisello, 2017), near-infrared fillers and pigments (Qin, 2015; Xie et al., 2019), reflective fillers (TiO_2 & SiO_2) (Chen et al., 2019; Jiang et al., 2020; Yi et al., 2019), nano titanium dioxide (TiO_2) with waterborne polyurethane as binder (Miao et al., 2022) are being implemented for yielding heat-reflective coating to reduce the pavement surface performance. The decrease in pavement surface temperature is recognized as a form of passive cooling. Retro-reflective materials, alternatively termed directionally reflective coatings, possess the ability to reflect solar radiation in the same direction as the incoming rays. This characteristic allows for the redirection of solar heat beyond the urban canyon. Pisello (2017) demonstrated a notable reduction in pavement surface temperature of up to 8.5 °C. Additionally, the air temperature at a height of 1.8 m above the ground experienced a decrease of up to 1.7 °C. These findings highlight the considerable cooling impact as well as the potential of RR materials as effective tools in urban heat management strategies.

Near-infrared pigments and fillers have been developed for coating pavements, with the aim of reducing the reflection of visible light while enhancing the reflection of near-infrared light when compared to conventional light-coloured pavements. Generally, pavement thermal environments are primarily influenced by solar radiation, which spans wavelengths from 300 to 3000 nm. This spectrum can be categorized into distinct regions based on wavelength, including ultraviolet radiation (below 400 nm), visible radiation (400–700 nm), and near-infrared radiation (700–2500 nm). The near-infrared region radiation significantly impacts the thermal conditions of the environment, making it a crucial factor to consider in urban heat effect mitigation. By using the pigments and fillers that operate in the near-infrared spectrum at optimal coating formula (B(Red iron oxide):C(R-TiO_2):A(NIR-TiO_2) = 5:25:8) in heat-reflective coating, the pavement surface temperature was reduced by about 9 °C as compared to the control asphalt pavement. Similarly, Qin (2015) determined that the high reflectance of near-infrared paint managed to reduce the pavement surface temperature for about 8 °C–20 °C based on their initial albedo value.

Numerous researchers have implemented reflective materials such as rutile titanium dioxide (TiO_2) with a high refractive index, as the reflective filler in the production of heat-reflective coatings. The inclusion content of TiO_2 plays a crucial role in impacting the temperature reduction. For instance, in a study conducted by Chen et al. (2019), various proportions of reflective at 4.5%, 9.0%, 13.5% and 18.0% were tested, resulting in a maximum pavement surface reduction in pavement surface temperature of 8.18 °C. Similarly, Yi et al. (2019) experimented with different functional filler/base material ratios, including 5:5, 4:6, and 6:4. The most effective cooling effect was observed with the ratio at 5:5, leading to the highest reduction in surface temperature by 8 °C. In a parallel study, Jiang et al. (2020) also identified the optimal ratio of functional materials to base materials as 5:5 and determined that the most effective dosage of the coating was 0.6 kg/m^2, resulting in a significant reduction in surface temperature. Other than that, Miao et al. (2022) also utilized TiO_2 as the reflective filler in conjunction with waterborne polyurethane (WPU) serving as the binder to unify all components. The notable feature of WPU is its exceptional cooling capacity, capable of absorbing heat within a temperature range of 2 °C–105 °C. This attribute enhances the solar reflectivity on the pavement surface, leading to a substantial reduction in pavement surface temperature by 10.2 °C. The combination of TiO_2 and WPU highlights the synergistic effect of using reflective fillers in tandem with specialized binders to achieve superior heat-reflective coatings for pavements.

Briefly, global case studies indicate that implementing cooling pavement strategies can reduce the micro-level effects of UHI to enhance thermal comfort. Although there are a few subtle differences in cooling efficacy between the methods of application outlined above, they are all extremely competitive technologies for applying heat-reflective asphalt pavement. Appropriate technology should be selected to construct a heat-reflective pavement by considering the environmental conditions.

3.2.3.2. Assessing climate impact via surrounding air temperature reduction. The implementation of heat-reflective pavements also plays a role in modifying the urban cooling effect and mitigating climate impacts by lowering pavement surface temperatures. In a study conducted in California, Mohegh et al. (2017) utilized the weather research and forecasting model (WRF) with a 4 km spatial resolution to simulate the current climate conditions. Through this simulation, they monitored reductions in air temperature in cities with widespread deployment of cool pavements. The researchers reported an overall mean bias in surface air temperature for the entire state of California at −0.30 °C, achieved by increasing the albedo within the range of 0.1–0.4. The annual average surface air temperatures in selected California cities showed reductions ranging from 0.18 °C to 0.86 °C, with peak temperature reductions observed in late morning and evening, particularly when boundary layer heights were low and solar irradiance (late morning) or heat accumulation in the pavement (evening) was high. Additionally, an online coupled WRF-urban modelling system was employed to assess the regional impact of pavements on metropolitan Phoenix, considering land-atmosphere interactions and land surface heterogeneity (Yang et al., 2016). The 10-d simulation demonstrated that reflective pavements, with increased albedo ranging from 0.17 to 0.5, resulted in a temperature reduction of approximately 0.5 °C. However, this temperature reduction was deemed limited in high-temperature regions such as Phoenix, where the pavement surface temperature at 1400 local time often exceeds 45 °C. In summary, reflective pavements have the potential to decrease regional air temperatures.

The transition from traditional pavements to cool pavement designs exhibited slight decreases in air temperature, ranging from 33 °C to approximately 32 °C at critical points (Giorio and Paparella, 2023). Notably, areas with low sky view factor (SVF), like narrow urban canyons, experienced a reduction in air temperature of up to 0.3 °C, while more significant decreases were observed in canyons with an east-west orientation and high SVF (Croce et al., 2021). Pavement replacement, especially with cool pavements, contributed to reported reductions in air temperature within the range of 0.15 °C–3.0 °C, aligning with findings from previous studies (Qin, 2015; Yi et al., 2019). At an altitude of 20 m, a decrease in daytime temperatures from 0.2 °C to 0.4 °C was evident, with a further reduction to 0.2 °C at an altitude of 30 m during the daytime and 0.1 °C during nighttime hours. This decrease in air temperature was attributed to the cooling effect of cool pavements, contributing to thermal comfort and stability in the urban environment.

3.2.3.3. Carbon emissions. Human activities, particularly the burning of fossil fuels for transportation and energy production, are the primary drivers of carbon emissions, leading to the accumulation of greenhouse gases like carbon dioxide (CO_2) in the atmosphere. These gases trap heat and contribute to global warming. Consequently, urban areas, characterized by dense infrastructure, dark surfaces, and limited vegetation, absorb, and retain more heat, exacerbating the UHI effect. However, by employing advanced materials engineered to deflect sunlight and reduce heat absorption, heat-reflective pavement not only reduces the demand for energy-intensive cooling systems in cities but also directly mitigates carbon emissions associated with such processes. This dual benefit not only significantly decreases carbon footprints but also helps to cool urban environments by combating the UHI effect.

According to findings by Yang et al. (2016), a net increase in the albedo of urban areas by about 0.1 could result in a negative radiative forcing on Earth equivalent to offsetting 4.4×10^4 Mt of CO_2 emissions. This offsetting effect was achieved by modifying pavement albedo, accounting for 40% (2.0×10^4 Mt), while the remaining 60% (2.4×10^4 Mt) was attained using construction materials with reflective properties. Based on Yang et al. (2016) research, they employed an index named daily effective albedo, which is defined as the ratio between total reflected shortwave radiation and total incoming shortwave radiation over an urban canyon across the diurnal cycle. Throughout the year, the implementation of reflective pavement resulted in a significant increase in pavement albedo, measured at 0.15. This increase corresponded to elevated daily effective albedo levels, with values of 0.070, 0.045, and 0.021. The rise in average daily effective albedo translated to a proportional increase in Earth's albedo, ranging from 4.2×10^{-5} to 1.4×10^{-4}. These shifts in albedo corresponded to changes in radiative forcing, estimated from 5.33 to 17.78 MW, with a potential one-time global CO_2 offset of approximately 2780–9330 Mt. Within urban canyons, reflective pavements demonstrated the ability to offset CO_2 emissions equivalent to approximately 13.9%–46.6% of emissions without building canopies.

In the United States, especially in highly populated states such as New York, California, or Texas, the CO_2 emissions each year is recorded around 40 million metric tons (MMT) by mostly originating from residential heating and cooling (Roxon et al., 2020). Implementing a cool pavement approach such as heat-reflective pavement could lead to a CO_2 equivalent reduction ranging from 0.9 to 3.1 MMT in Boston across 50 years. This emission cut is similar to the removal of 5500 to 17,800 vehicles' worth of emissions for the same time span. For the larger city of Phoenix, estimated savings are between 2.2 and 16.7 MMT of CO_2 equivalent over a 50-year period, which is equivalent to emissions from 13,700 to 96,000 vehicles. Considering Boston's pledge to reach carbon neutrality by 2050 (Boston Environment Department, 2020) and Phoenix's goal for net-zero emissions by 2060 (City of Phoenix, 2009), cool pavement strategies could account for 1.0%–3.0% and 0.7%–6.0% reductions in their respective overall greenhouse gas emissions over the next 50 years. Although the percentages may be perceived as modest, they signify substantial potential emissions reduction that can be achieved with minor additional cost by selecting suitable pavement materials during normal maintenance periods. Across the global urban expanse of nearly 4 billion km^2 (CIESIN, 2016), with around 7.5% comprising paved surfaces, the proper selection of pavement materials could contribute to a global CO_2 equivalent emission decrease in the magnitude of 5.0 gigatons (GT) to 44.7 GT over the approaching 50 years.

3.2.3.4. Feasibility and profitability analysis of technology efficacy. The technology readiness level (TRL) serves as a criterion for the cost-effectiveness analysis. TRL is an evaluation system used to assess the maturity level of a particular technology, and the TRL rating is assigned based on the project's development (Tzinis, 2015). It is ranked between 1 and 9, with TRL 1 denoting recognition and clarification of only the most fundamental concepts. TRL 2 occurs when the fundamental principles have been investigated, and practical applications can be applied, but the technology lacks experimental proof of concept. When the research is initiated, the technology will progress to TRL 3. A proof-of-concept model is generally required and constructed at this stage. TRL 4 is achieved once the proof-of-concept technology is complete, followed by TRL 5 that is identified as a breadboard technology and requires more extensive testing. A TRL 6 technology has a prototype or model that is entirely functional. In order to qualify as TRL 7 technology, the operational model or prototype must be demonstrated in space. Once the technology has been assessed and deemed suitable for integration into an existing technology system during TRL 8, the technology can be implemented. TRL 9 indicates that the technology has been extensively developed, tested, and demonstrated during a mission with success. Fig. 14 shows the TRL definition based on the early NASA model.

Based on all 43 research articles, the TRL and cooling performance of each type of heat-reflective asphalt pavement was evaluated. Table 5 displays the maturity level of the cool pavement technologies. The findings demonstrate that the modification of asphalt pavement materials is still in the early stage of experimental development. In contrast, the application of heat-reflective coating and pavement treatment incorporating cool materials has reached a level of technological maturity where they can be used in real-world situations and environments. Therefore, the heat-reflective coating is one of the best methods to be applied in cool pavement based on its efficiency and level of readiness.

For pavement construction, the expenses associated with transporting the pavement materials are of great concern to the relevant authorities. Due to the high cost of transporting aggregate, Middel et al. (2020) asserted that the price of pavement aggregate is contingent on local availability. The price of aggregate is anticipated to be increased by about 50%, which adds about 12% to the original cost of resurfacing and about 2% to the original cost of repaving when the aggregates are required to be transported from other areas. Table 6 presents a list of various pavement types and their respective solar reflectance values. The listed cost is solely pertaining to repaving a pre-existing thoroughfare under municipal jurisdiction within the Sacramento locality. From the findings, it is observed that light-coloured materials are priced similarly to the conventional darker-coloured materials over a short-term of time. However, light-coloured materials, such as white asphalt shingles, may be more resistant to the deteriorating effects of thermal stress and solar radiation and last longer. White asphalt shingles are less absorbent than dark asphalt shingles, even minor variations in albedo can have a significant effect on service life. Therefore, long-life road pavements require less maintenance, thereby reducing future expenditures for road administrations and restorations (Gáspár and Bencze, 2020). According to Gilbert et al. (2017), enhancing the albedo of a square metre of pavement in Los Angeles by 0.20 resulted in a passive annual cooling site energy savings of approximately 0.22 kWh/m^2. This, in turn, results in savings of approximately 2.9 MJ in conditioning primary energy demand (PED), and approximately $0.03 in conditioning energy cost. The utilization of

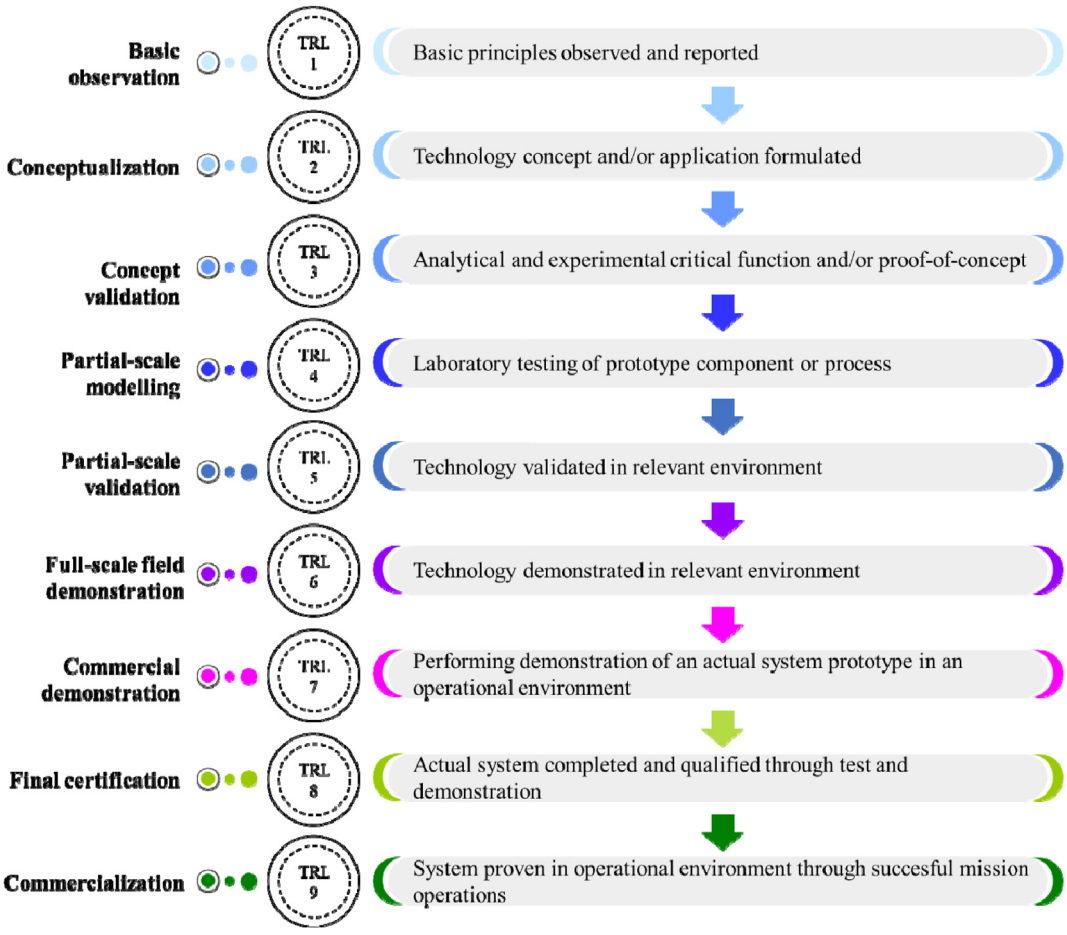


Fig. 14. Definition of TRL according to the early NASA model (Fasterholdt et al., 2018).

Table 5
Comparison of heat-reflective pavement technology based on TRL (Cheela et al., 2021).

Method of application	Cooling performance	TRL (1–9)
Heat-reflective coating	High	9
Modification of asphalt binder	Medium	4
Pavement treatment using cool materials	High	8
Modification of pavement materials	Medium	5

Table 6
Cost and albedo estimate for various pavements with updated lifetimes (Middel et al., 2020).

Pavement type	Service life (year)	Cost ($94/m^2)	Albedo (new) (%)	Albedo (weathered) (%)
Asphalt (18 cm)	15	18	5–10	15–20
Whitetopping (13 cm)	25	18	35–40	25–30
Asphalt with light aggregate (18 cm)	15	18	5–10	35–40
Whitetopping with white cement	25	21	70–80	40–60

materials that possess a higher solar reflectance could be a viable initiative that promotes efficient life cycle costing of pavements.

3.2.3.5. Cost analysis of heat-reflective pavements. Heat-reflective pavement holds great promise for enhancing urban liveability and sustainability by lowering surface temperature, alleviate heat-related issues and eventually mitigating UHI effects. However, while the environmental benefits of heat-reflective pavements are well-documented, questions regarding their economic viability, particularly concerning maintenance and manufacturing costs, remain paramount. Understanding the complexities and implications of cost maintenance and manufacturing cost is crucial for evaluating the overall feasibility and sustainability of integrating heat-reflective pavements into urban infrastructure. This introduction delves into the multifaceted considerations surrounding cost maintenance and manufacturing costs, exploring the challenges, opportunities, and implications associated with the economic aspects of heat-reflective pavement implementation. In the manufacturing process of heat-reflective pavement, the use of raw superior reflective materials such as titanium dioxide, diatomite, and hollow glass beads is essential. However, these materials come with a high initial cost, contributing to the overall manufacturing expenses (Chen et al., 2022). In Phoenix, a cool pavement reflective sealant costs approximately 14% more than a typical asphalt sealant, with typical asphalt sealants priced at around $4.40 per square yard and cool pavement reflective sealants priced at approximately $5 per square yard (Brian, 2021).

In addition, mitigating the urban heat island phenomenon not only improves urban liveability but also aids in lowering global temperatures. A slight reduction of 1 °C in average urban temperatures can yield considerable benefits, including an estimated annual cost saving of 15 million dollars from reduced air conditioning use (Brian, 2021). This highlights the economic and practical value of UHI mitigation strategies like employing heat-reflective pavements and emphasizes the importance of investing in sustainable urban infrastructure initiatives.

Previously, U.S. Environmental Protection Agency (EPA) has summarized a range of construction and maintenance costs for conventional and cool pavements as shown in Table 7. In comparison, pavements incorporating light-colored aggregates generally entail lower costs compared to alternative pavement modifications. As a result, the adoption of reflective pavement presents an economically favorable option for pavement projects seeking cost-effective solutions.

At times, achieving optimal performance and enhancing the cooling effect of the heat-reflective coating necessitates a higher dosage of reflective fillers, thereby leading to escalated costs. While this intensification effectively enhances the pavement's cooling capabilities, it brings about additional expenses, presenting a notable challenge in the development and adoption of heat-reflective pavement solutions (Chen et al., 2022). Consequently, striking a balance between the need for effective cooling and the associated expenses remains a crucial consideration in advancing the feasibility and widespread implementation of heat-reflective pavement technologies.

4. Moving forward

4.1. Review gaps

The literature has identified several gaps in cool pavement technologies. First and foremost, numerous studies exist on the cooling performance of the heat-reflective pavement. However, the available study on the durability of the heat-reflective asphalt pavement is limited. The cost-effectiveness of heat-reflective pavement is not well determined by previous researchers. In addition, experimental studies confirmed the energy-saving characteristic of heat-reflective pavement. However, there is a need for more studies on the enduring performance of this pavement in terms of its durability.

Previously, there were primarily four methods in constructing the heat-reflective pavement. Thermal reflective coating is one of the most efficient and widely used methods. In recent decades, numerous studies on its cooling performance have been conducted. However, technological advancement in this field is extremely sluggish, as the study trend has remained relatively constant over the past few years. Researchers have utilized nearly identical materials and methodologies in developing cool pavements. Therefore, it is essential to research new technologies in this field in this era of modern technology to enhance the pavement performance and mitigate the drawbacks of heat-reflective asphalt pavement.

In addition, there is also a research gap in the field of life cycle cost analysis for heat-reflective pavement lies in the integration of comprehensive life cycle assessments with cost evaluations pertaining to the overall lifespan of pavements. This entails examining not only the initial construction costs but also the operational and maintenance expenses incurred over the pavement's entire lifespan. By incorporating such assessments, researchers can optimize proposed solutions in alignment with EU directives aimed at achieving carbon neutrality. This holistic approach ensures that the chosen pavement solutions not only mitigate the urban heat island effect but also contribute to sustainability goals by minimizing carbon emissions throughout their lifecycle.

4.2. Limitations

The SLR employed inclusion and exclusion criteria after the search strategy, excluding more than half of the articles, despite developing a comprehensive search string. Therefore, the number of articles included is lower than anticipated, with only 43 articles discussing recent studies on heat-reflective asphalt pavement. Even though the included paper demonstrates a consistent research format and is highly pertinent to the requirements of the research question, it could be enhanced by utilizing multiple databases to find more relevant articles and conducting a convincing and exhaustive review. This review focuses primarily on the capacity of heat reflectance on the pavement to mitigate UHI, so the article sources are primarily from the engineering disciplines. Rather than comprehending the mechanism of the system, most of the researchers in this field identify their knowledge of heat-reflective asphalt pavement based on an experiment or previous relevant research.

The protocol for this review is a compilation of a variety of well-established methods that are frequently and extensively employed in biomedical research. Modifications were made to the biomedical strategy to make it compatible with engineering research. The risk of bias and repeatability of the source of the data extracted for this review's analysis are evaluated to assure the quality of this review. When incomplete information is encountered during data extraction, assumptions and other supplementary sources are employed to resolve the issue. In addition, the readers should interpret the information carefully as the discussed research question is investigated only based on the 43 included articles.

4.3. Recommendations

Upon thorough examination of the selected studies, the authors have put forth several recommendations for future scholars to consider when incorporating reflective coatings as a means of mitigating urban heat island (UHI) effects in urban design. To effectively integrate reflective coatings into urban landscapes, careful attention must be paid to regional climatic conditions, seasonal variations, and the unique characteristics of the urban microclimate. The selection of materials and their cooling impact is intricately tied to ambient circumstances, necessitating further investigation into the thermal characteristics and resilience of reflective coatings at both laboratory and field scales. Expanding the repertoire of materials by incorporating waste and industrial byproducts into paving materials not only benefits the environment but also reduces manufacturing costs, contributing to improved waste management practices.

The existing focus of studies on short-term performance in asphalt mixtures underscores the need for extensive research on the long-term mechanical performance, durability, and thermal properties of modified pavements to ensure sustained cooling effects over time. A more detailed analysis of thermal conductivity, heat capacity, and

Table 7
Cost comparison for various pavements (EPA, 2017).

Project category	Pavement type	Modification approach	Approx. installation cost ($/sqft)	Projected lifespan (year)
New construction	Asphalt pavement	Hot mix asphalt (HMA) using light-coloured aggregates sourced locally	0.10–1.50	7–20
	Concrete pavement	Portland cement with plain joints	0.30–4.50	15–35
	Non-vegetated permeable pavement	Porous asphalt	2.00–2.50	7–10
	Vegetated permeable pavement	Pervious concrete	5.00–6.25	15–20
		Paving blocks	5.00–10.00	>20
		Grass/gravel pavement	1.50–5.75	>10
Maintenance	Surface application	Chip seals with light-coloured aggregates sourced locally	0.10–0.15	2–8
		Micro-surfacing	0.35–0.65	7–10
		Ultra-thin white topping	1.50–6.50	10–15

permeability of materials is required, with optimal dosage recommended to be less than 5%–6%. Further exploration of the use of binder modifiers to enhance low-temperature efficacy is warranted, along with an examination of long-term pavement albedo fluctuations, considering aging effects and utilizing climate modelling to comprehend albedo changes.

In the quest for a comprehensive understanding, researchers should delve into the correlation between aggregate porosity and the mechanical properties of asphalt mixtures. Aging phenomena in asphalt pavements coated with cool materials should be a focal point in future studies, encompassing the impact of repeated illumination on coating optical characteristics. The development of finite element models and thermal computations is essential to quantify energy savings. Additionally, conducting experiments to explore variations in daily, seasonal, and yearly patterns related to urban design and climate conditions is imperative. Lastly, a deeper exploration of the chemical reactions between materials is necessary to ensure adequate adhesion strength within asphalt mixtures for long-term service.

5. Conclusions

In this SLR, the critical importance of heat-reflective asphalt pavements in mitigating the UHI effect is thoroughly examined and underscored. By synthesizing a vast array of research findings, the review sheds light on the pressing need to adopt innovative cooling technologies to effectively tackle the challenges arising from rapid urbanization and climate change. Through meticulous screening and analysis of existing literature, the review brings into focus the imperative for further investigation into the resilience and efficacy of heat-reflective pavements. It emphasizes their potential to significantly contribute to environmental sustainability efforts and mitigate the adverse impacts associated with rising urban temperatures. By providing a comprehensive assessment of current knowledge and research gaps, this review serves as a valuable resource for policymakers, urban planners, and stakeholders involved in decision-making processes. It lays a solid foundation for informed policy formulation and strategic interventions aimed at fostering resilient and sustainable urban environments.

Based on the systematic reviews performed, the authors have reviewed and defined the parameters and criteria of each type of cool pavement technology to qualify the most effective and relevant technology for reducing UHI. This review covers only the technology of heat-reflective asphalt pavement, as opposed to evaporative or heat storage-modified pavement, even though there are numerous other technologies for performing cool pavement. Utilizing a heat-reflective coating with optimum coating brushing quantities on the pavement's surface exhibits the highest mean cooling efficacy (10.97 °C) compared to other methods. Moreover, titanium dioxide (TiO_2) exhibits advantageous characteristics including a high refractive index, significant light scattering capacity, effective UV shielding, robust stability, oxidation resistance, high reflectance of near-infrared radiation, and substantial blocking of solar thermal radiation. These properties position TiO_2 as a promising candidate for serving as the primary functional filler material to enhance the reflectivity of heat-reflective coatings. Incorporating TiO_2 particles has demonstrated significant effectiveness, with the cooling effect increasing by 80% as the content was elevated from 2% to 14%. Furthermore, the integration of heat-reflecting asphalt, in conjunction with thermal insulation coatings and void-rich asphalt pavements, offers multifaceted mechanisms for mitigating the urban heat island effect. These approaches not only improve solar reflectance but also enhance the mechanical performances, including fatigue strength, binder stiffness, and rutting resistance, of asphalt pavement.

Enhancement of cooling effect by heat reflective pavement in mitigating UHI effect promotes increment in albedo value. The changes in albedo value effectively lowering surface temperatures and minimizing energy consumption for cooling, these pavements demonstrate significant environmental and economic benefits. Heat-reflective pavements offer a promising solution to mitigate urban heat island effects and enhance urban sustainability. While challenges such as initial manufacturing costs and maintenance expenses exist, the long-term advantages, including improved pavement durability and substantial reductions in carbon emissions, underscore the importance of investing in heat-reflective pavement technologies. With ongoing research and technological advancements, heat-reflective pavements stand poised to play a pivotal role in creating more liveable and resilient urban environments in the future. In a nutshell, while significant strides have been made in the study of heat-reflective pavements, there remain critical gaps in understanding durability, long-term performance, and life cycle cost analysis. Future research should focus on innovative technologies, comprehensive life cycle assessments, and addressing biases to advance sustainable urban infrastructure and mitigate urban heat island effects effectively.

Declaration of competing interest

The authors do not have any conflict of interest with other entities or researchers.

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